生物统计和生物信息引论

Lecture 2

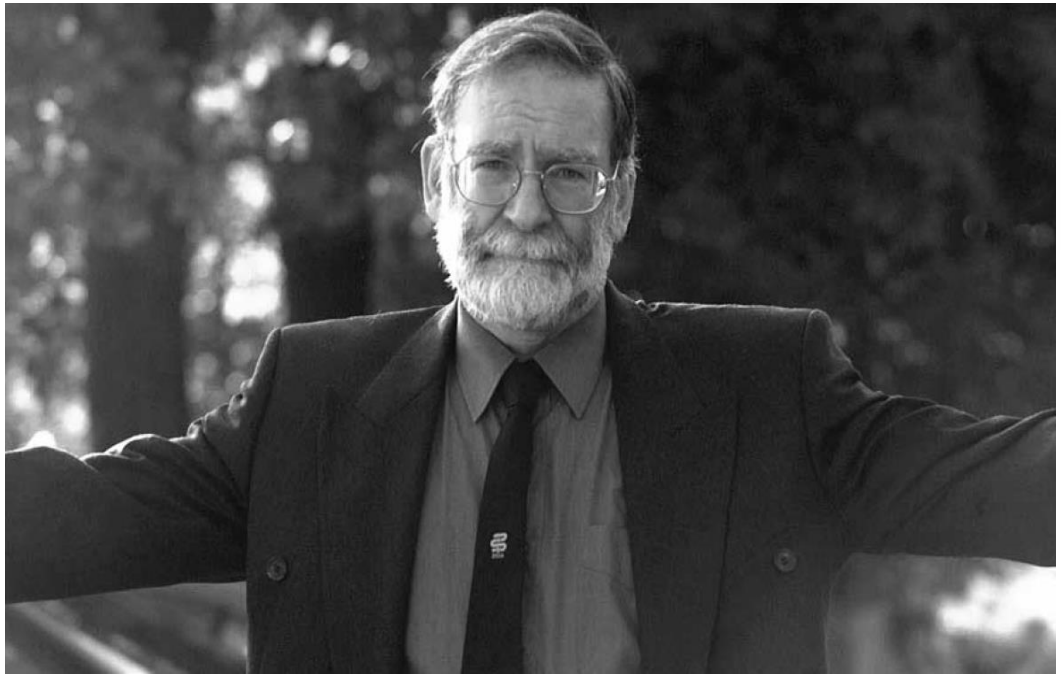
统计推断基础回顾

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Case Study: Harold Shipman

Dr. Harold Shipman was a British GP who became the UK's most prolific serial killer, murdering at least 215 elderly patients.

He was ultimately exposed when analysts noticed abnormal patterns in the timing of deaths and unusually high mortality linked to his practice.



Could Harold Shipman have been caught earlier?

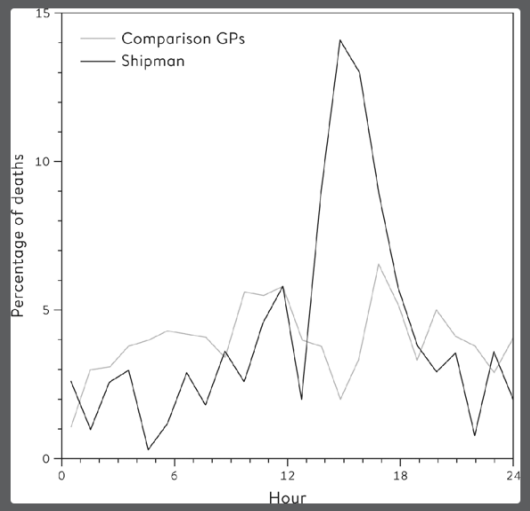


Figure 0.2
The time at which Harold Shipman's patients died, compared to the times at which patients of other local general practitioners died. The pattern does not require sophisticated statistical analysis.

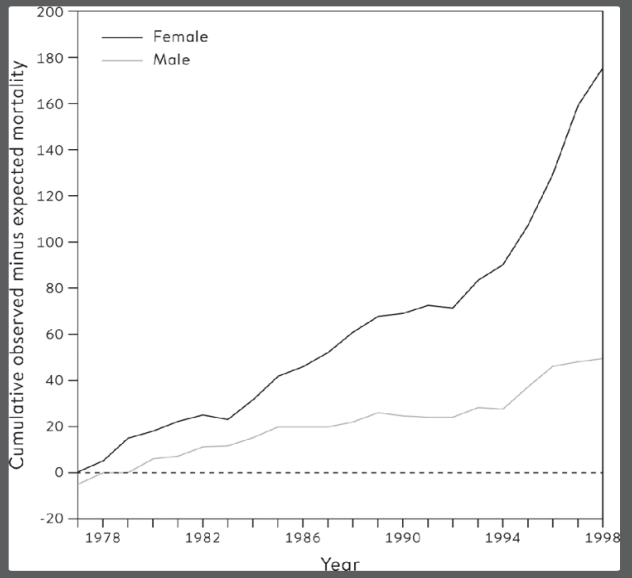


Figure 10.3
Cumulative number of death certificates signed by Shipman for patients who were age 65 or over and who died at home or in his practice. The expected number, given the composition of his practice list, has been subtracted.

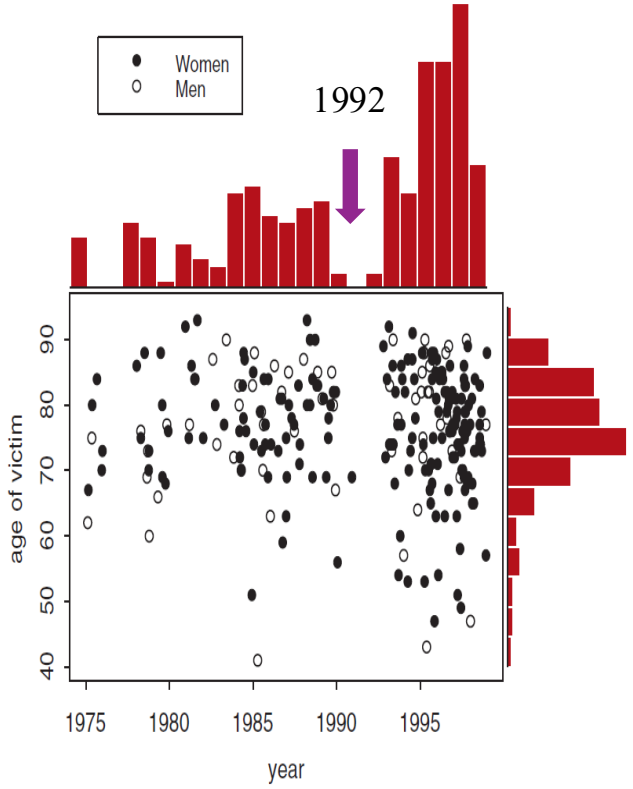


Figure 1. Age and sex of Shipman's victims by date of death

By 1979, the p-value for Shipman's excess mortality reached **0.004**.

The Gap**:** Monitoring 25,000 GPs meant that such a p-value was not unusual given the scale of testing.

This highlights the critical challenge of **Multiple Testing**.

Population vs. Sample

- **Population:**
  - Biologists: a defined group of humans or of another species of organisms.
  - Statisticians: a group of measurements about which one wishes to draw conclusion.
- **Sample:** A representative subset drawn from the population.
- **Random Sampling:**
  - Each member of the population has an equal chance of being selected.
  - Independent: the selection of any member of the population must in no way influence the selection of any other member.

"We study the sample to understand the population."

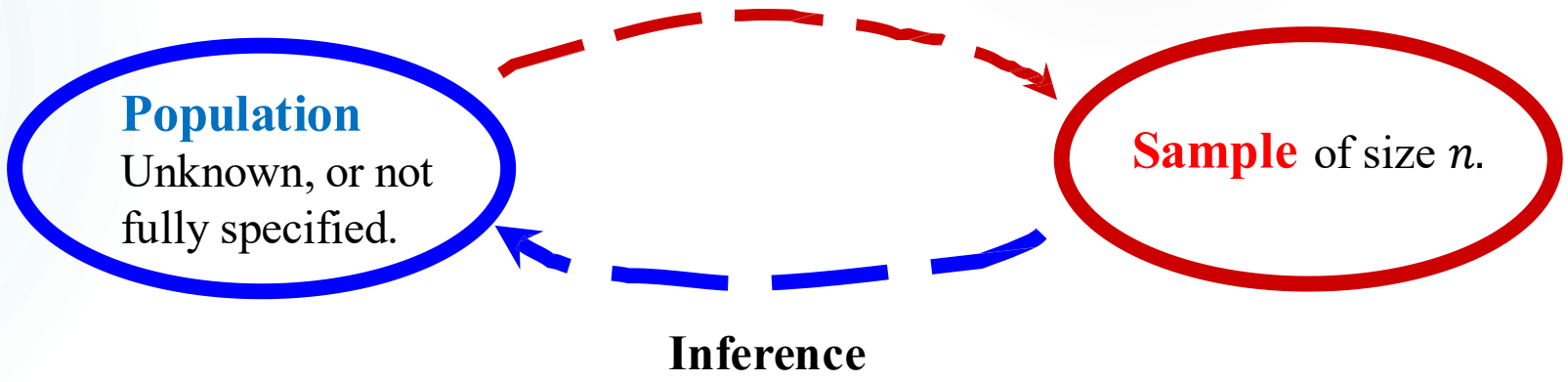
Population vs. Sample

Population = all elements of interest Characterized by a distribution F with some parameter θ

Sample = the data $X_1, \ldots, X_n$, selected subset of the population

- Population is usually characterized by a r.v. X, or a distribution function, or a density function

Examples of F: Bernoulli(p), Normal(μ, σ), Gamma(n, λ), Poisson(λ), etc.



Parameters vs. Statistics

Parameters

Numerical characteristics of a **Population**.

- Mean, Variance

μ, σ^2

Statistics

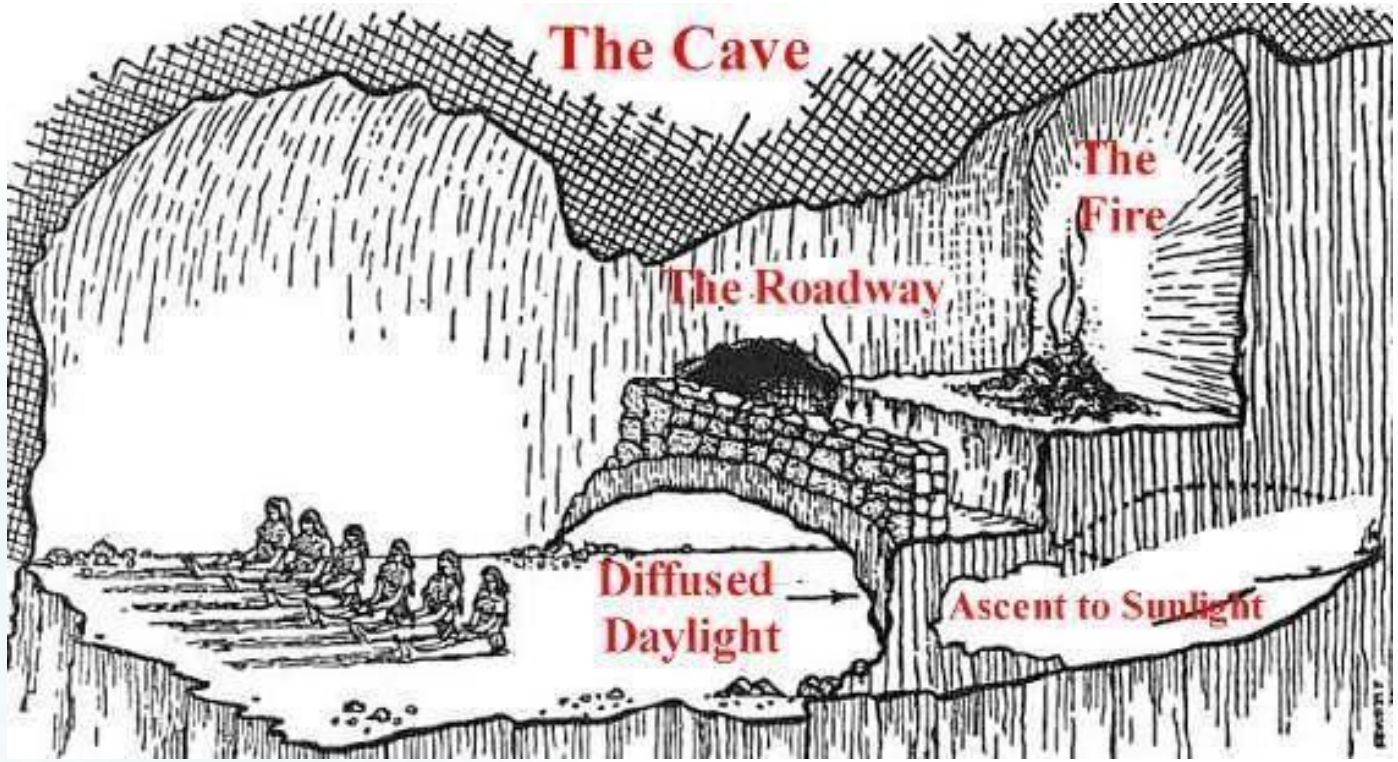
Numerical characteristics of a **Sample**.

- An estimate of a population parameter
- Varies from sample to sample

$\bar{Y}, S^2$

- **Statistical inference**: inference about the population based on a sample.
  - **Parametric Inference**: An inference about the unknown parameter(s) of the specified distribution. (For instance, we assume that the data are from $N(\mu, \sigma^2)$, where μ and σ^2 are unknown)
  - **Non-Parametric Inference**: An inference about the function (e.g., mean, variance, and distribution function) of the population distribution.

Statistical allegory of the cave



Two Different Types of Procedures

- **Estimating population parameters**
  - Point estimation

    Using a sample statistic to estimate a population parameter.
  - Interval estimation

    Estimation of the amount of variability in a sample statistic when many samples are repeatedly taken from a population.
- **Hypothesis testing**
  - The comparison of sample results with a known or hypothesized population parameter.

Independence

Definition (Discrete Case) Two random variables X and Y are **independent** if $P(X = x, Y = y) = P(X = x)P(Y = y)$, $\forall x, y$.

Definition (Continuous Case) Two random variables X and Y are **independent** if $f_{X,Y}(x, y) = f_X(x)f_Y(y)$, $\forall x, y$.

Equivalent Formulation For any mearuable functions g and h, $E\{g(X)h(Y)\} = E\{g(X)\}E\{h(Y)\}$.

Basic Properties If X and Y are independent, $E(XY) = E(X)E(Y)$, $\text{cov}(X, Y) = 0$, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$, and $g(X)$ and $h(Y)$ are independent.

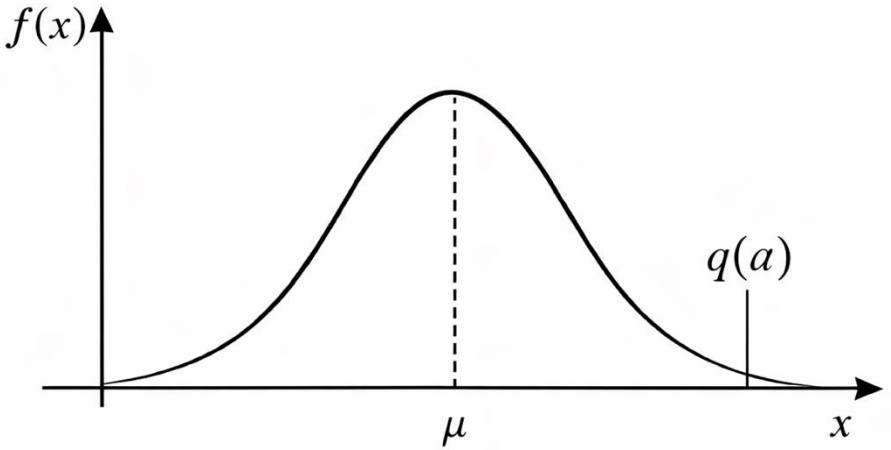
Independent and Identically Distributed (i.i.d.) A sequence of ranfom variables $X_1, \ldots, X_n$ is called independent and identically distributed (i.i.d.) if

1. Independence. $f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$.
2. Identical distribution. $X_1, \ldots, X_n \sim F$, i.e., all variables share the same distribution: $F_{X_1}(x) = \ldots = F_{X_n}(x) = F(x)$.

Normal and Multivariate Normal Distribution

- **Definition.** A random variable X follows a **Normal distribution** if $X \sim N(\mu, \sigma^2)$, with probability density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$



- μ is the mean and σ^2 is the variance.
- **Definition.** A random variable $\boldsymbol{X} \in \mathbb{R}^p$ follows a **multivariate Normal distribution** if $\boldsymbol{X} \sim N(\boldsymbol{\mu}, \Sigma)$, with probability density

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}$$

- $\boldsymbol{\mu} \in \mathbb{R}^p$ is the mean vector, and $\Sigma \in \mathbb{R}^{p\times p}$ is the covariance matrix (symmetric and positive definite).

Marginal distribution Suppose we partition $\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}$, $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$. Then $\boldsymbol{X}_1 \sim N(\boldsymbol{\mu}_1, \Sigma_{11})$.

Conditional distribution A key property: $\boldsymbol{X}_1 \mid \boldsymbol{X}_2 = \boldsymbol{x}_2 \sim N(\boldsymbol{\mu}_{1|2}, \Sigma_{1|2}$, where $\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\boldsymbol{x}_2 - \boldsymbol{\mu}_2)$, $\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

Chi-Square Distribution

- **Definition.** (Chi-Square Distribution) Let $X_1, X_2, \ldots, X_r \ \ i.i.d. \sim N(0,1)$. Then the distribution of the random variable

$$\xi = \sum_{i=1}^{r} X_i^2,$$

is known as the *Chi-square* distribution with r *degrees of freedom*, and it is denoted as

$$\xi \sim \chi_r^2.$$

- **Density.**

$$f(x) = \frac{1}{\Gamma\left(\frac{r}{2}\right)2^{r/2}} x^{(r/2)-1} e^{-x/2}, \qquad x > 0, \qquad \Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y}\, dy, \qquad \alpha > 0.$$

Example: Sample variance of Normal distribution; likelihood ratio test statistics; Goodness-of-fit: Pearson χ^2 test (testing whether the sample is from a given distribution)

Chi-Square Distribution

- This distribution is used in certain statistical inference problems involving **confidence intervals for variances** and **testing hypotheses about variances**

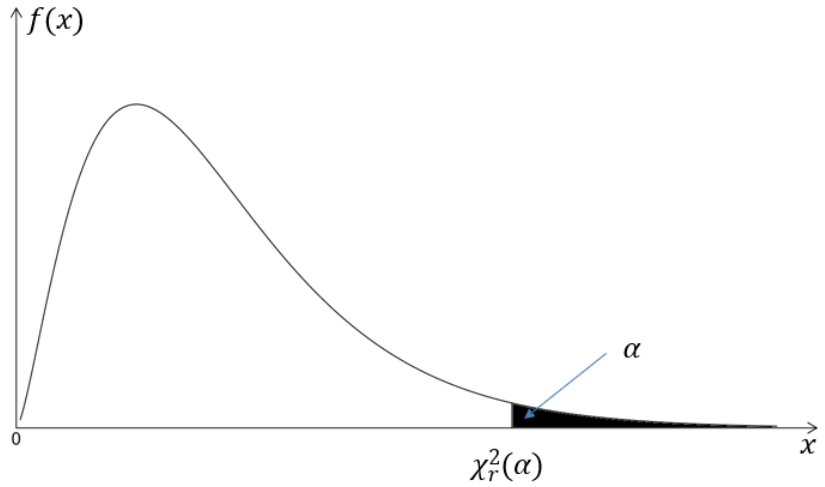
- **Property.** Let $\xi \sim \chi^2$, then

(1) The c.f. of ξ is $\varphi(t) = (1-2it)^{-n/2}$;

(2) $E\xi = r$, and $Var(\xi) = 2r$;

(3) Let $\xi_1 \sim \chi^2_{r_1}$, $\xi_2 \sim \chi^2_{r_2}$ and ξ_1, ξ_2 are independent, then $\xi_1 + \xi_2 \sim \chi^2_{r_1+r_2}$.

(4) Let $\xi_i \sim \chi^2_{r_i}$, $i = 1, 2, \cdots, k$ and $\xi_1, \xi_2, \cdots, \xi_k$ are independent, then $\sum_{i=1}^{k} \xi_i \sim \chi^2_{r_1+\cdots+r_k}$.



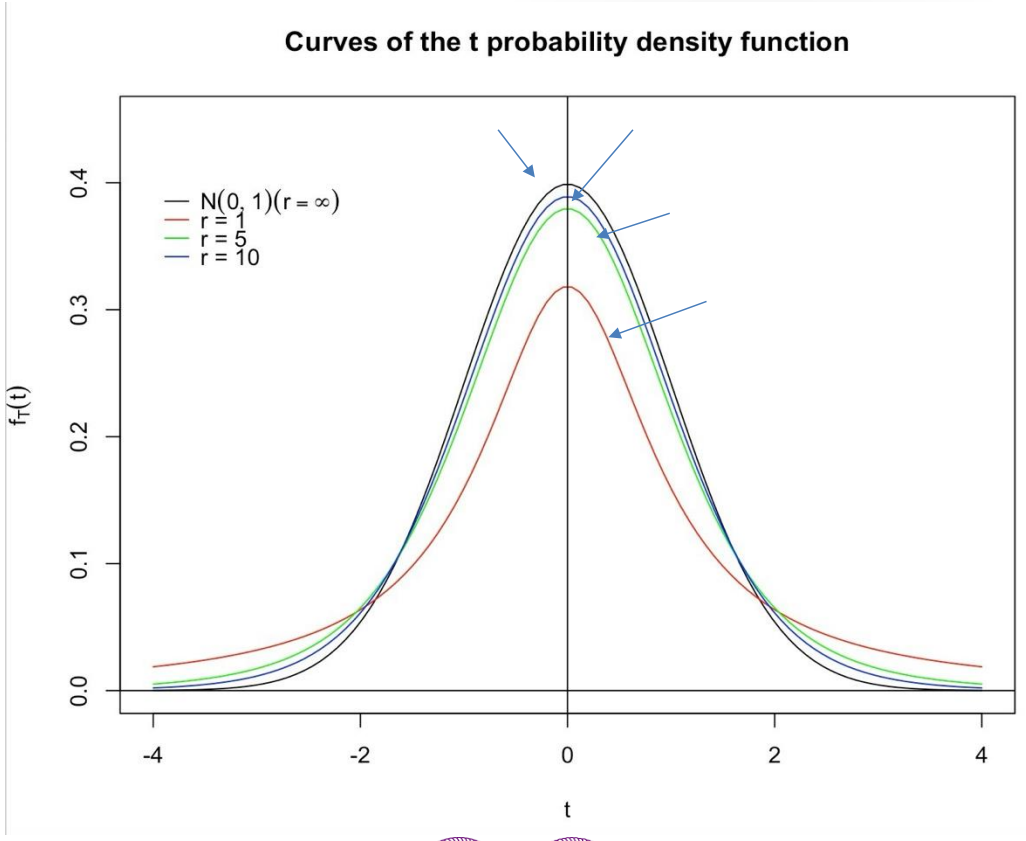
- **Upper α Quantile $\chi^2_r(\alpha)$:** The upper α quantile $\chi^2_r(\alpha)$ of Chi-Square distribution is defined as the constant c such that $P(\xi > c) = \alpha, 0 < \alpha < 1$.

t Distribution

- **Definition.** (t Distribution) Let X and Y be two independent r.v.'s distributed as follows:$X \sim N(0,1)$ and $Y \sim \chi_r^2$, and define the r.v. T by: $T = \frac{X}{\sqrt{(Y/r)}}$. The r.v. T is said to have the (Student's) t-*distribution* with r *degrees of freedom*. The notation used is: $T \sim t_r$.

- **Density.**

$$f_T(t) = \frac{\Gamma\left(\frac{r+1}{2}\right)}{\Gamma\left(\frac{r}{2}\right)\sqrt{\pi r}}\left(1+\frac{t^2}{r}\right)^{-(r+1)/2}, \quad t \in R$$



t Distribution

- **Property.**

(1) If r.v. $T \sim t_r$, then $E(T^k)$ exits only if $k < r$ $(r > 1)$ and

$$E(T^k) = \begin{cases} r^{\frac{k}{2}} \frac{\Gamma\left(\frac{k+1}{2}\right)\Gamma\left(\frac{r-k}{2}\right)}{\Gamma\left(\frac{r}{2}\right)\Gamma\left(\frac{1}{2}\right)}, & k \text{ is even}, \\ 0, & k \text{ is odd}. \end{cases}$$

In particular, when $r \geq 2$, $E(T) = 0$ and when $r \geq 3$, $Var(T) = r/(r-2)$;

(2) When $r = 1$, t_1 is the Cauchy distribution and its p.d.f. is

$$f_1(t) = \frac{1}{\pi(1+t^2)}, \quad -\infty < t < \infty;$$

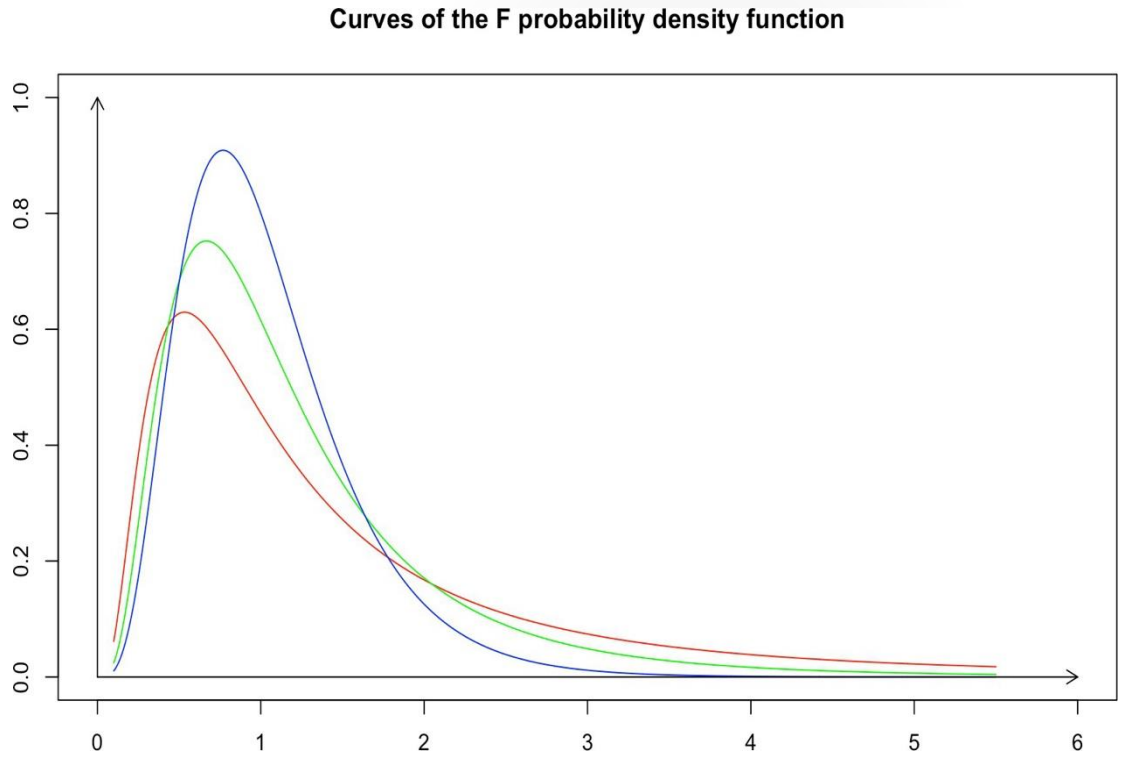
(3) As $r \to \infty$, t_r converges to $N(0, 1)$.

F Distribution

- **Definition.** (F Distribution) Let X and Y be two independent r.v.'s distributed as follows: $X \sim \chi^2_{r_1}$ and $Y \sim \chi^2_{r_2}$, and define the r.v. F by: $F = \frac{X/r_1}{Y/r_2}$. The r.v. F is said to have the *F-distribution* with r_1 and r_2 *degrees of freedom*. The notation often used is: $F \sim F_{r_1,r_2}$.
- **Density.**

$$f_{r_1,r_2}(x) = \begin{cases} \frac{\Gamma\left(\frac{r_1+r_2}{2}\right)}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)} \left(\frac{r_1}{r_2}\right)^{\frac{r_1}{2}} x^{\frac{r_1}{2}-1} \left(1+\frac{r_1}{r_2}x\right)^{-\frac{r_1+r_2}{2}}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Example: Hypothesis testing for variance of two population



F Distribution

▶ Property.

(1) If $F \sim F_{r_1,r_2}$, then $1/F \sim F_{r_2,r_1}$;

(2) If $T \sim t_r$, then $T^2 \sim F_{1,r}$;

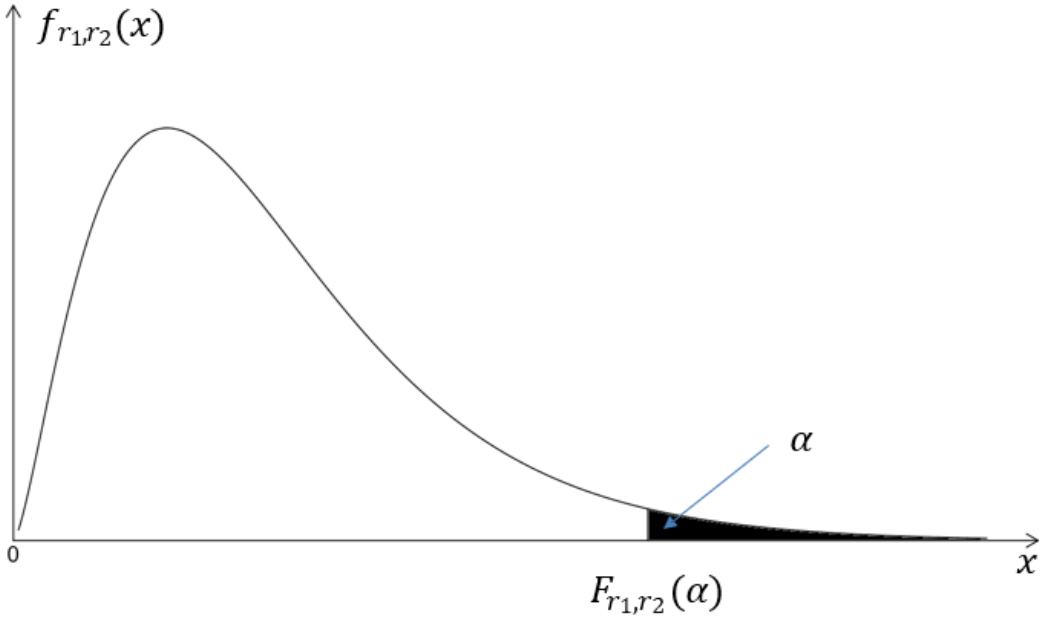
(3) If $F \sim F_{r_1,r_2}$, then for $k > 0$ and $2k < r_2$,

$$E(F^k) = \left(\frac{r_2}{r_1}\right)^k \frac{\Gamma\left(\frac{r_1}{2}+k\right)\Gamma\left(\frac{r_2}{2}-k\right)}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)}.$$

In particular,

$$E(F) = \frac{r_2}{r_2-2},\ r_2 \geq 3, \quad \text{and} \quad Var(F) = \frac{2r_2^2(r_1+r_2-2)}{r_1(r_2-2)^2(r_2-4)},\ r_2 \geq 5;$$

(4) $F_{r_1,r_2}(1-\alpha) = 1/F_{r_2,r_1}(\alpha)$.



- **Upper α Quantile $F_{r_1,r_2}(\alpha)$:** The upper α quantile $F_{r_1,r_2}(\alpha)$ of the F distribution is defined as the constant c such that $P(F > c) = \alpha, 0 < \alpha < 1$.

Exponential Family

- **Definition.** (Exponential family). Let $\mathcal{F} = \{ f(x,\theta) : \theta \in \Theta \}$ be a distribution family, where Θ is the parameter space. If the p.d.f. $f(x,\theta)$ has the following form:

$$f(x,\theta) = C(\theta) \cdot \exp\left\{ \sum_{i=1}^{k} Q_i(\theta) T_i(x) \right\} \cdot h(x),$$

where k is a positive integer, $C(\theta) \geq 0$ and $Q_i(\theta)$ are functions of θ (they cannot depend on x), $h(x) \geq 0$ and $T_i(x)$ are functions of x (they cannot depend on θ), then $\mathcal{F}$ is said to be an *exponential family*.

Example: Normal distribution, Gamma distribution, Binomial distribution, Poisson distribution.

- **Natural (Canonical) Form** If the exponential family has the following form:

$$f(x,\theta) = C^*(\theta) \cdot \exp\left\{ \sum_{i=1}^{k} \theta_i T_i(x) \right\} \cdot h(x),$$

then it is said to be the natural (canonical) form. The parameter space

$$\Theta^* = \{ (\theta_1, \theta_2, \ldots, \theta_k) : \int_{\mathcal{X}} \exp\left\{ \sum_{i=1}^{k} \theta_i T_i(x) \right\} h(x)\, dx < \infty \},$$

is said to be the natural parametric space.

Point Estimation

- **Definition.** If we **use a statistic to ESTIMATE an unknown parameter(s),** then it is called **an (point) estimator of the parameter**.

- The typical (point) **estimators** for μ and σ^2 are

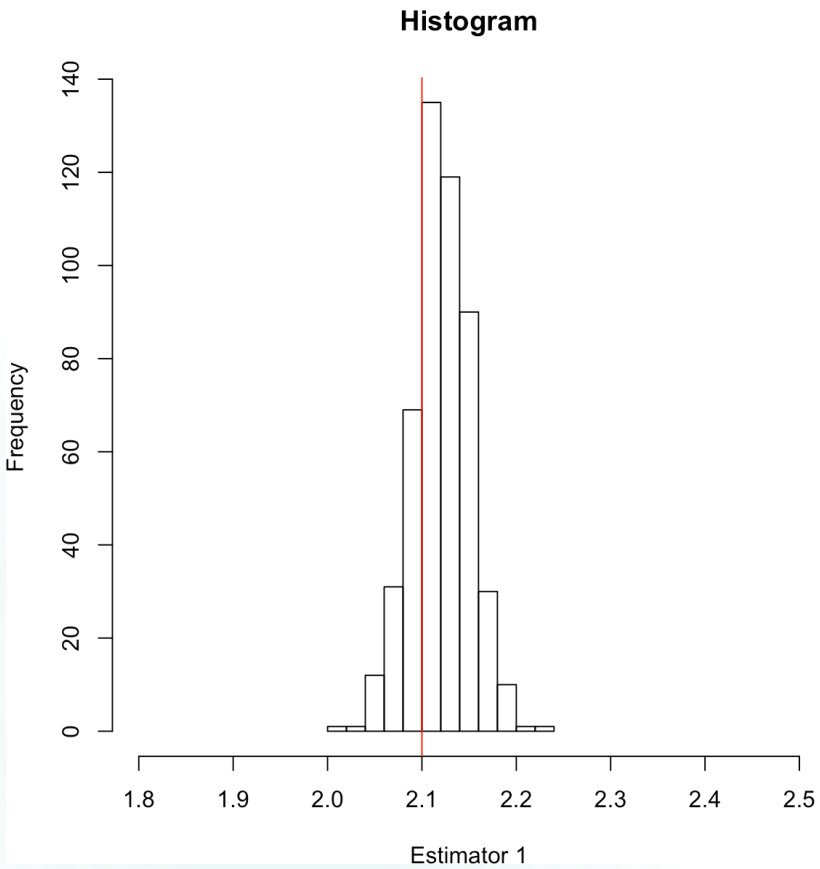
$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

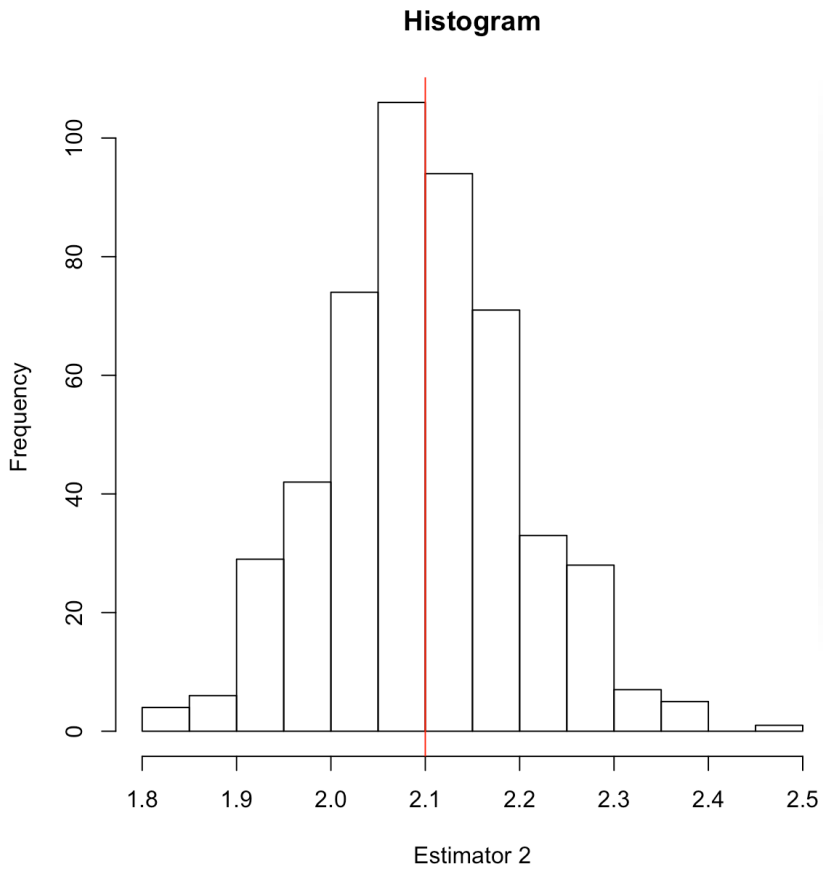
- Remark that in statistics, the observed value of the estimator is called an **estimate** of the unknown parameter.

- For instance, $\bar{x}$ and s^2 are the respective estimates of μ and σ^2

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \text{and} \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

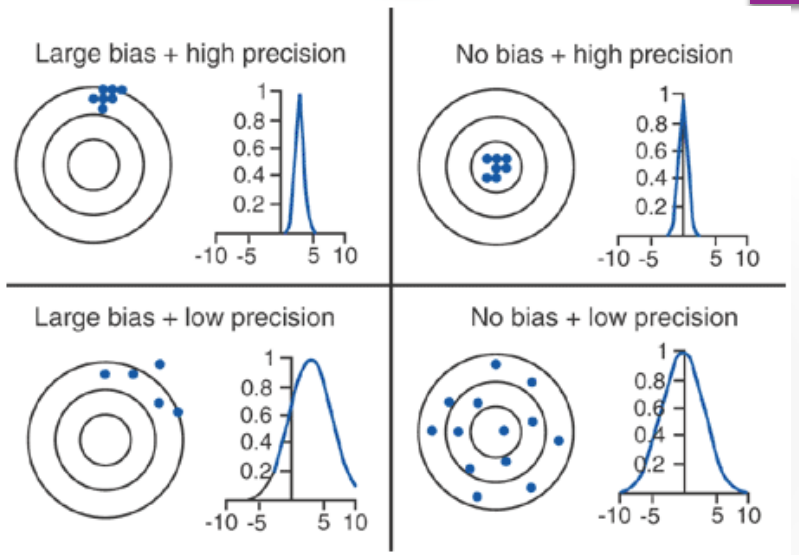
Which Estimator is Better?

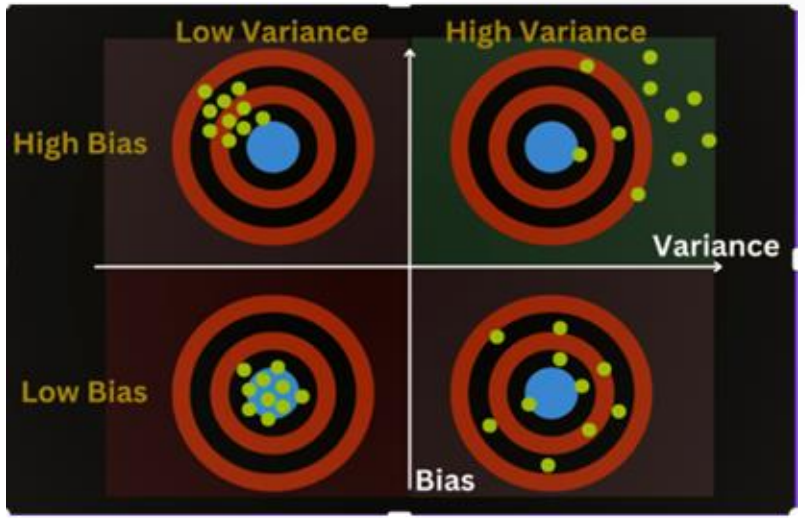




Estimator Quality: Bias & Precision

- What is a good estimate?
  - Unbiased
  - Precision
  - Consistent
- When you want to estimate a treatment effect
  - Mean tells only half the story.
  - **Variability** of estimated treatment effect determines the precision and reproducibility of the estimation.
- Manage Variability
  - Larger n → smaller SE → higher precision.
  - But n is constrained by budget and resources.
  - Good design anticipates and controls sources of variation.
  - Methods can adjust for and separate known variation.





Mean Squared Error (MSE)

- **Bias** of an estimator: $E(\hat{\theta}) - \theta$
- **Variance** of an estimator: $Var(\hat{\theta})$
- The goodness of an estimator can be measured as $|\hat{\theta} - \theta|$ (mean absolute error $E|\hat{\theta} - \theta|$, is a reasonable alternative)
- Any increasing function of $|\hat{\theta} - \theta|$ would serve to measure the goodness of an estimator. In particular, $(\hat{\theta} - \theta)^2$ can be served as this purpose.
- **Definition.** The mean squared error (MSE; 均方误差) of an estimator $\hat{\theta}$ of a parameter θ is the function of θ defined by $E_\theta(\hat{\theta} - \theta)^2$.

Mean Squared Error (MSE)

- MSE measures the average squared difference between the estimator $\hat{\theta}$ and the parameter θ.

- MSE has at least two advantages over other distance measures.
  - It is quite tractable analytically
  - It has the following interpretation

$$E_\theta(\hat{\theta}-\theta)^2 = Var_\theta(\hat{\theta}) + (E_\theta\hat{\theta}-\theta)^2 = Var_\theta(\hat{\theta}) + [Bias_\theta(\hat{\theta})]^2$$

MSE incorporates two components, one measuring the variability of the estimator (precision) and the other measuring its bias (accuracy). An estimator that has good MSE properties has smaller combined variance and bias.

Maximum Likelihood Estimation (MLE)

- **Likelihood**: $L(\theta; x) = f(x; \theta)$
- **Log-likelihood**: $l(\theta; x) = \log(L)$
- **Definition.**(MLE) Let $\boldsymbol{X} = (X_1, \ldots, X_n)$ be a random sample from the distribution family $\boldsymbol{\mathcal{F}} = \{ f(x; \theta), \theta \in \Theta \}$ where $f(x; \theta)$ is the p.d.f., and let $\boldsymbol{x} = (x_1, \ldots, x_n)$ be the respective observed values. The likelihood function, $L(\theta; \boldsymbol{x})$, is given by $L(\theta; \boldsymbol{x}) = \prod_{i=1}^{n} f(x_i; \theta)$. The log-likelihood function, $l(\theta; \boldsymbol{x})$, is given by $l(\theta; \boldsymbol{x}) = \log L(\theta; \boldsymbol{x}) = \sum_{i=1}^{n} \log \sum f(x_i; \theta)$. Let $\hat{\theta} = \hat{\theta}(\boldsymbol{X})$ be an estimator of θ satisfying

$$L(\hat{\theta}; \boldsymbol{x}) = \sup_{\theta \in \Theta} L(\theta; \boldsymbol{x}), \qquad \boldsymbol{x} \in \boldsymbol{\mathcal{X}},$$

or equivalently,

$$l(\hat{\theta}; \boldsymbol{x}) = \sup_{\theta \in \Theta} l(\theta; \boldsymbol{x}), \qquad \boldsymbol{x} \in \boldsymbol{\mathcal{X}}.$$

Then $\hat{\theta} = \hat{\theta}(\boldsymbol{X})$ is called a *maximum likelihood estimator (estimate)* of θ, and $\hat{\theta}(\boldsymbol{X})$ is called a *maximum likelihood estimate* of θ.

MLE in an Exponential Family

- Let $\boldsymbol{X} = (X_1, \ldots, X_n)$ be a random sample from an exponential family with p.d.f.

$$f(x;\theta) = C(\theta)\exp\left\{\sum_{i=1}^{k}\theta_i T_i(x)\right\}h(x), \quad \theta \in \Theta,$$

where Θ is the natural parameter space. Let Θ_0 be the set of inner points of Θ. The likelihood function of $\boldsymbol{X}$ is

$$L(\theta;\boldsymbol{x}) = \prod_{j=1}^{n} f(x_j;\theta) = C^n(\theta)\exp\left\{\sum_{i=1}^{k}\theta_i \sum_{j=1}^{n} T_i(x_j)\right\}h(\boldsymbol{x}),$$

where $h(x) = \prod^{n}_\{i = 1\}\, h(x_i)$. The log-likelihood function is

$$l(\theta;\boldsymbol{x}) = \log L(\theta,\boldsymbol{x}) = n\log C(\theta) + \sum_{i=1}^{k}\theta_i \sum_{j=1}^{n} T_i(x_j) + \log h(\boldsymbol{x}).$$

- **Theorem.** If for any random sample $\boldsymbol{X} = (X^1, \ldots, X_n)$, the likelihood equations

$$\frac{n}{C(\theta)}\frac{\partial C(\theta)}{\partial \theta_i} = -\sum_{j=1}^{n} T_i(x_j), \ i = 1, \cdots, k$$

has a solution in Θ_0, then the solution is unique and is the MLE of θ.

Reference: 陈希孺. 数理统计引论. 北京: 科学出版社, 1981, 1998. (Theorem 2.6.3 on Page 199)

Properties of ML Estimators

Large Sample Properties.

- **Consistent**: $\forall \varepsilon > 0, \quad \lim_{n\to\infty} \Pr\left(\left|\hat{\theta}_n^{ML} - \theta\right| > \varepsilon\right) = 0.$
- **Asymptotical normality**: $\hat{\theta}^{ML} \overset{a}{\sim} N[\theta, \{I(\theta)\}^{-1}]$
- Variance-Covariance is the Rao-Cramer lower bound (if the model is well-specified) – **efficient**
- **Invariance**: If θ^{ML} is the ML estimator of θ, then $\gamma^{ML} = g(\theta^{ML})$ is the maximum likelihood estimator of $\gamma = g(\theta)$. This means that rather than estimating a parameter θ, we can instead estimate some function of it, $g(\theta)$. We can then recover an estimate of θ from $g(\theta)$.

Thus, ML is best asymptotically normal.

Small Sample Properties.

We don't know much about MLE's small sample properties. In general, though, if there is a good small sample estimator, it will look like maximum likelihood. Note that maximum likelihood may not be unbiased in small samples.

Information Matrix and Standard Errors

- The **variance of an ML estimator**, $\hat{\theta}^{ML}$ is calculated by the **inverse** of the Information matrix:

$$\text{var}(\theta) = [I(\theta)]^{-1}$$

- **Information matrix** is the negative of the expected value of the **Hessian matrix**:

$$[I(\theta)] = -E[H(\theta)]$$

- The **Hessian** is the matrix of second derivatives of the likelihood with respect to the parameters:

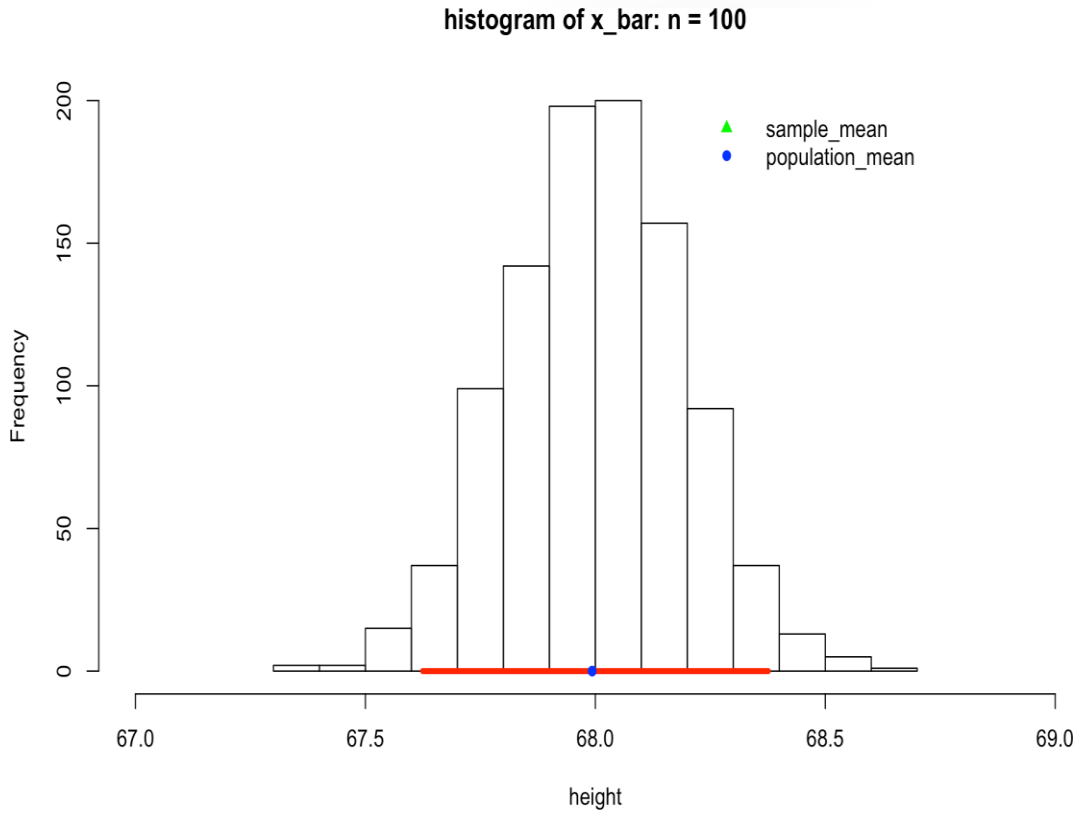
$$H(\theta) = \frac{\partial^2 \ln \mathcal{L}(\theta)}{\partial \theta \partial \theta'}$$

- Thus, the variance-covariance matrix of $\hat{\theta}^{ML}$ is:

$$\begin{aligned} \text{var}(\theta) &= [I(\theta)]^{-1} \\ &= (-E[H(\theta)])^{-1} \\ &= \left(-E\left[\frac{\partial^2 \ln \mathcal{L}(\theta)}{\partial \theta \partial \theta'}\right]\right)^{-1} \end{aligned}$$

Confidence Interval (CI)

- Point estimate $\hat{\theta}$:
  - *Drawback*: only a single point, do not directly know its precision (or uncertainty; risk); random sample is just one of numerous possible random samples, and thus point estimate is just one of numerous plausible estimates
  - *MSE*: not a direct evaluation
- Any statistics $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_1 < \hat{\theta}_2, [\hat{\theta}_1 - \hat{\theta}_2]$ is an interval estimate of θ (also called **confidence interval**).



Confidence Interval (CI)

Definition 1. Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample with p.d.f. $f(x;\theta)$, $\theta \in \Theta \subseteq R$. Then,

- *random interval*: an interval whose end-points are r.v.'s;
- *confidence level*: the probability of the confidence interval $[\hat{\theta}_1, \hat{\theta}_2]$ containing θ, i.e., $P_\theta(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2)$;

- Reliability and precision cannot be obtained simultaneously
- Given a level of reliability, find an interval with the highest precision above the level -- Neyman

Method to Construct CI – Pivot Variable Method

- How to construct CI? For example, the 95% CI of mean height of children of age 18 at Hong Kong?

Example 1. Let $X_1, \ldots, X_n$ be a random interval from the $N(\mu, \sigma^2)$ distribution, where only one of μ or σ^2 is unknown. Construct a confidence interval for it with confidence coefficient $1 - \alpha$.

Discussion.

- *Let μ be unknown and σ^2 be known.* The natural point estimate of μ is the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ whose distribution is $\bar{X} \sim N(\mu, \sigma^2/n)$. Standardization,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1),$$

whose distribution does not depend on μ. Let $z_{\alpha/2}$ be the upper $\alpha/2$ quantile, then

$$P_\mu\left(\left|\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}\right| \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Method to Construct CI – Pivot Variable Method

$$P_\mu\left(\left|\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}\right| \le z_{\alpha/2}\right) = 1-\alpha.$$

That is,

$$P_\mu\left(\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \le \mu \le \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) = 1-\alpha.$$

Therefore, $[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}]$ is a confidence interval for μ with confidence level $1-\alpha$.

- What's the method used in example 1?

(a) Think of a r.v. which contains the parameter θ, the r.v.'s $X_1, \ldots, X_n$, preferably in the form of a sufficient statistic, and whose distribution is (exactly or at least approximately) known.

(b) Determine suitable points $a < b$ such that the r.v. in step (a) lies in $[a, b]$ with P_θ-probability $\ge 1-\alpha$.

(c) In the expression of step (b), rearrange the terms to arrive at an interval with the end-points being statistics and containing θ.

(d) The interval in step (c) is the required confidence interval.

Method to Construct CI – Pivot Variable Method

Let σ^2 be unknown and μ be known. Let $S_\mu^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2$ and recall that

$$\frac{nS_\mu^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi_n^2.$$

The r.v. nS_μ^2/σ^2 satisfies the requirements of step (a). From the Chi-square tables, determine any pair $0 < a < b$ for which $P(a \le X \le b) = 1 - \alpha$, where $X \sim \chi_n^2$. Then

$$P_{\sigma^2}\left(a \le \frac{nS_\mu^2}{\sigma^2} \le b\right) = 1 - \alpha, \quad \text{for all } \sigma^2, \quad \text{or}$$

$$P_{\sigma^2}\left(\frac{nS_\mu^2}{b} \le \sigma^2 \le \frac{nS_\mu^2}{a}\right) = 1-\alpha, \quad \text{for all } \sigma^2 \text{ and steps (b) and (c) are satisfied.}$$

In particular (for simplicity, but not optimal),

$$P_{\sigma^2}\left(\frac{nS_\mu^2}{\chi^2_{n;\alpha/2}} \le \sigma^2 \le \frac{nS_\mu^2}{\chi^2_{n;1-\alpha/2}}\right) = 1 - \alpha, \quad \text{for all } \sigma^2.$$

It follows that

$$\left[\frac{nS_\mu^2}{\chi^2_{n;\alpha/2}}, \frac{nS_\mu^2}{\chi^2_{n;1-\alpha/2}}\right], \quad S_\mu^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2$$

Method to Construct CI – Pivot Variable Method

Example 4. Let $X_1, \ldots, X_n$ be a random sample from the $N(\mu, \sigma^2)$ distribution, where both μ and σ^2 are unknown. Construct confidence intervals for μ and σ^2, each with confidence coefficient $1 - \alpha$.

Discussion. We have that:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1) \quad \text{and} \quad \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n} \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi^2_{n-1},$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$, and these two r.v.'s are independent. It follows that $\frac{\sqrt{n}(\bar{X}-\mu)/\sigma}{\sqrt{(n-1)S^2/\sigma^2(n-1)}} = \frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t_{n-1}$. From the t-tables, determine any pair $(a,\ b)$ with $a < b$ such that $P(a \le X \le b) = 1 - \alpha$, where $X \sim t_{n-1}$. It follows that:

$$P_{\boldsymbol{\theta}}\left[a \le \frac{\sqrt{n}(\bar{X} - \mu)}{S} \le b \right] = 1 - \alpha, \quad \text{for all } \boldsymbol{\theta} = (\mu, \sigma^2),$$

or

$$P_{\boldsymbol{\theta}}\left(\bar{X} - b\frac{S}{\sqrt{n}} \le \mu \le \bar{X} - a\frac{S}{\sqrt{n}} \right) = 1 - \alpha, \quad \text{for all } \boldsymbol{\theta}.$$

Method to Construct CI – Pivot Variable Method

In particular,

$$P_{\boldsymbol{\theta}}\left(\bar{X}-t_{n-1;\alpha/2}\frac{S}{\sqrt{n}}\le\mu\le\bar{X}+t_{n-1;\alpha/2}\frac{S}{\sqrt{n}}\right)=1-\alpha,\quad\text{for all }\boldsymbol{\theta},$$

where $P(X\ge t_{n-1;\alpha/2})=\frac{\alpha}{2}$ (and $X\sim t_{n-1}$). It follows that the required confidence interval for μ is:

$$\left[\bar{X}-t_{n-1;\alpha/2}\frac{S}{\sqrt{n}},\ \bar{X}+t_{n-1;\alpha/2}\frac{S}{\sqrt{n}}\right]=\bar{X}\pm t_{n-1;\alpha/2}\frac{S}{\sqrt{n}}\quad\text{(for brevity).}\tag{4}$$

The construction of a confidence interval for σ^2 in the presence of (an unknown) μ is easier. We have already mentioned that $\frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{n-1}$. Then repeat the process in Example 1(ii), replacing χ^2_n by χ^2_{n-1}, to obtain the confidence interval.

$$\left[\frac{(n-1)S^2}{\chi^2_{n-1;\alpha/2}},\ \frac{(n-1)S^2}{\chi^2_{n-1;1-\alpha/2}}\right],\quad S^2=\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2.\tag{5}$$

Hypothesis Testing

- **Parameter Estimation** and **Hypothesis Testing** are two main tasks of statistical inference.

Parametric hypothesis testing

- The form of d.f. (or p.d.f.) is known
- Test whether the parameter of interest lies or does not lie in a specified subset of the parameter space

Nonparametric hypothesis testing

- The distribution of the population is unknown
- The test methods should not depend on the mathematical formulation of the population distribution
- E.g., test whether the sample is drawn from Normal distribution

Hypothesis Testing

- **Null and alternative hypothesis**

$$H_0 : \theta \in \Theta_0 \longleftrightarrow H_1 : \theta \in \Theta_1.$$

- **Critical or rejection region** (拒绝域) (should not include unknown parameters)

$$D = \{\boldsymbol{X} = (X_1, \cdots, X_n) : |\bar{X} - 1| > A\}$$

- **Acceptance region** (接受域)

$$D^c = \{\boldsymbol{X} = (X_1, \cdots, X_n) : |\bar{X} - 1| \leq A\}$$

- **Nonrandomized test and randomized test**

$$\varphi(\boldsymbol{X}) = \begin{cases} 1, & \text{if } |\bar{X} - 1| > A, \\ 0, & \text{if } |\bar{X} - 1| \leq A. \end{cases}$$

$$\varphi(\boldsymbol{X}) = \begin{cases} 1, & \text{if } T(\boldsymbol{X}) > c, \\ r, & \text{if } T(\boldsymbol{X}) = c, \\ 0, & \text{if } T(\boldsymbol{X}) < c. \end{cases}$$

Hypothesis Testing

- **The power function** of the test employed is defined by:

$$\pi_\varphi(\theta) = P_\theta(\text{of rejecting } H_0 \text{ by the test } \varphi) = E_\theta[\varphi(\boldsymbol{X})]$$

- **Nonrandomized test**

$$\pi_\varphi(\theta) = P_\theta(\boldsymbol{X} = (X_1, \cdots, X_n) \in D), \quad \theta \in \Theta$$

- **Randomized test**

$$\pi_\varphi(\theta) = E_\theta[\varphi(\boldsymbol{X})] = P\left(\sum_{i=1}^n X_i > c\right) + rP\left(\sum_{i=1}^n X_i = c\right)$$

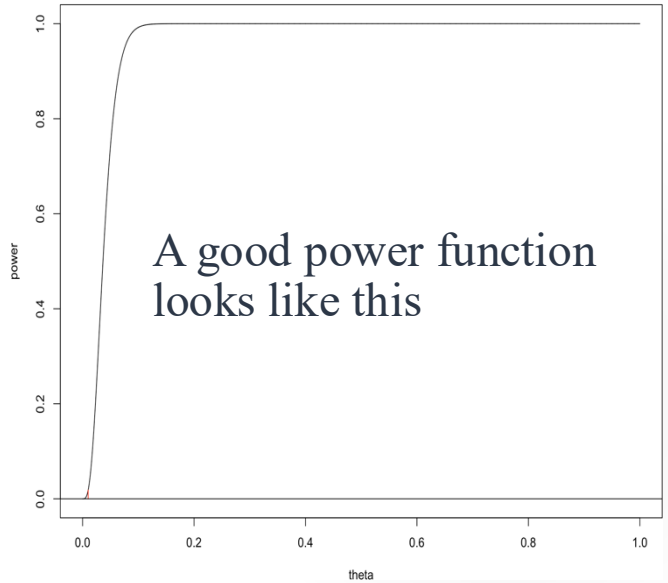
Hypothesis Testing

- **Type I error** can be expressed as

$$\alpha(\theta) = \begin{cases} \pi_\varphi(\theta), & \theta \in \Theta_0, \\ 0, & \theta \in \Theta_1. \end{cases}$$

- **Type II error** can be expressed as

$$\beta(\theta) = \begin{cases} 0, & \theta \in \Theta_0, \\ 1 - \pi_\varphi(\theta), & \theta \in \Theta_1. \end{cases}$$



Neyman-Pearson Principle: Choose a test (decision rule) such that

- first the probability of type I error is controlled at a preassigned level of significance α
- second the probability of type II error is made as small as possible (if it has maximum power among all tests with level of significance $\leq \alpha$)

Testing Parameters in a Single Normal Distribution

▶ Test for Normal Mean μ

Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$, consider the following hypothesis testing problem:

(1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

(2) $H_0 : \mu \leq \mu_0 \longleftrightarrow H_1 : \mu > \mu_0$,

(3) $H_0 : \mu \geq \mu_0 \longleftrightarrow H_1 : \mu < \mu_0$.

where μ_0 and the level of significance α is preassigned (or given).

Hypothesis testing problem (1) is called *two-side test*, while (2) and (3) is called *one-side test*.

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

Discussion. Obviously, $\bar{X} = \sum_{i=1}^{n} X_i/n$ is a good estimate of μ, and H_0 is more likely false if $|\bar{X} - \mu_0|$ is large. The rejection region can take the form of $D = \{\boldsymbol{X} = (X_1, \cdots, X_n) : |\bar{X} - \mu_0| > A\}$. To determine the critical value A, we know that when σ^2 is known, $\bar{X} \sim N(\mu_0, \sigma^2/n)$ under H_0. Standard $\bar{X}$, and when H_0 is true,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \sim N(0, 1).$$

We take U as test statistic, and the rejection region is

$$\{\boldsymbol{X} = (X_1, \cdots, X_n) : |Z| > c\}.$$

Since the level of significance is α, it holds that

$$P(|Z| > c \mid H_0) = P\left(\left|\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}\right| > c \mid H_0\right) = \alpha,$$

therefore, $c = z_{\alpha/2}$, the upper $\alpha/2$ quantile of $N(0, 1)$. The rejection region is

$$D = \left\{\boldsymbol{X} = (X_1, \cdots, X_n) : \left|\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}\right| > z_{\alpha/2}\right\}$$

Testing Parameters in a Single Normal Distribution

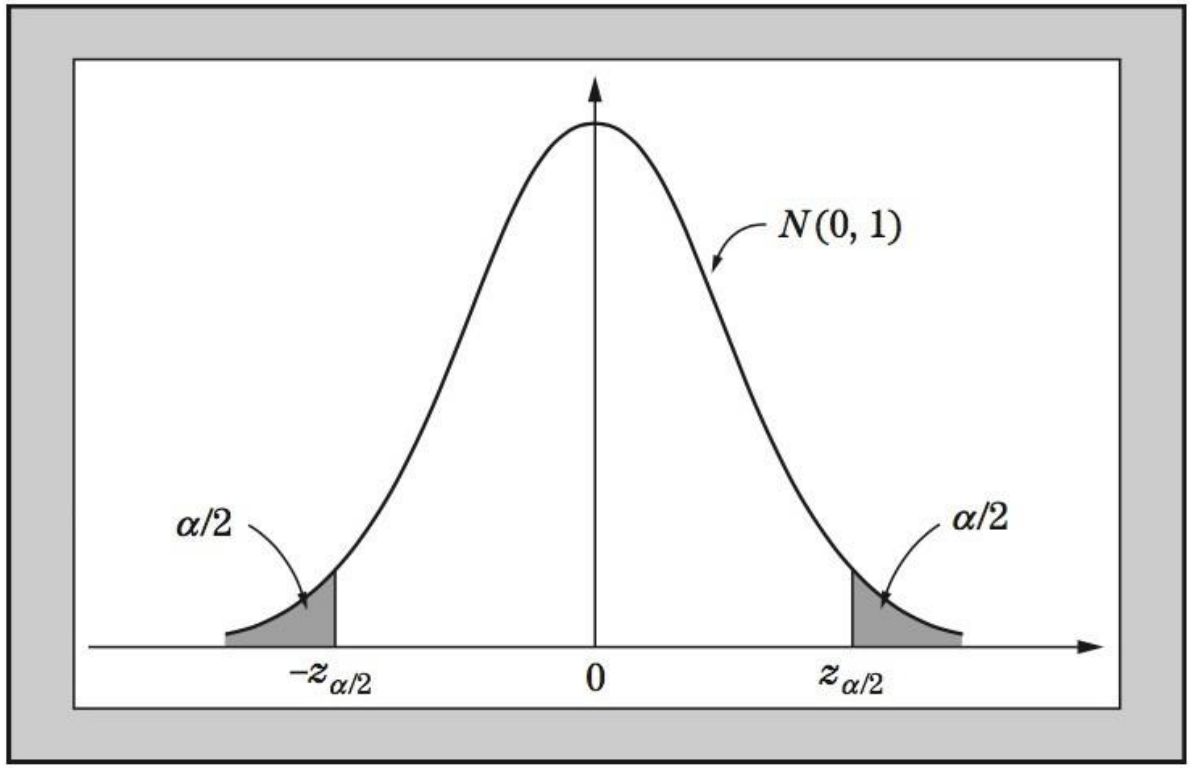


FIGURE 11.8

Rejection region of the hypothesis H_0 in (i) (the shaded areas), and the respective probabilities of rejection.

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

Let $\Phi(\cdot)$ be the d.f. of $N(0,1)$, the power function of the test is

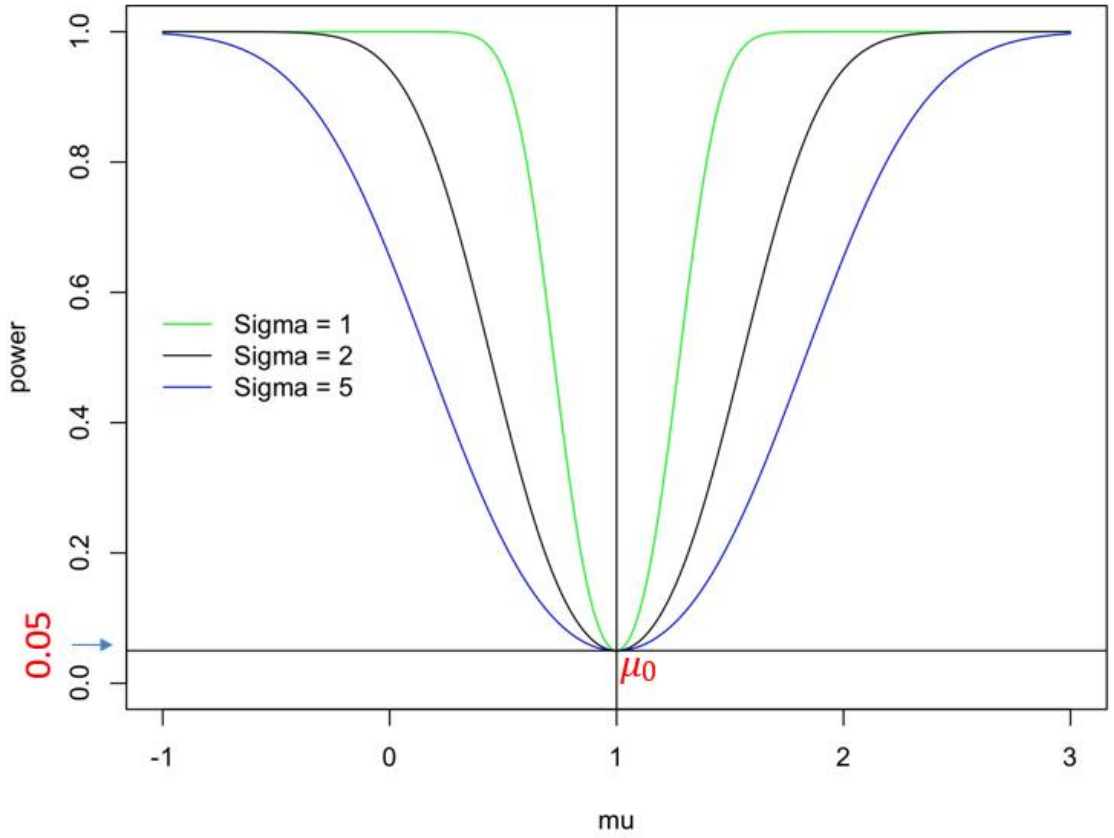
$$
\begin{aligned}
\pi_1(\mu) &= P_\mu\left(\left|\frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma}\right| > z_{\alpha/2}\right) \\
&= P_\mu\left(\left|\frac{\sqrt{n}(\bar{X}-\mu+\mu-\mu_0)}{\sigma}\right| > z_{\alpha/2}\right) \\
&= P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} > z_{\alpha/2}+\frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right) \\
&\quad + P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} < -z_{\alpha/2}+\frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right) \\
&= 1-\Phi\left(z_{\alpha/2}+\frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right)+\Phi\left(-z_{\alpha/2}+\frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right)
\end{aligned}
$$

Properties: (1) $\pi_1(\mu) = \pi_1(2\mu_0 - \mu)$; (2)) $\pi_1(\mu)$ is increasing when $\mu \geq \mu_0$

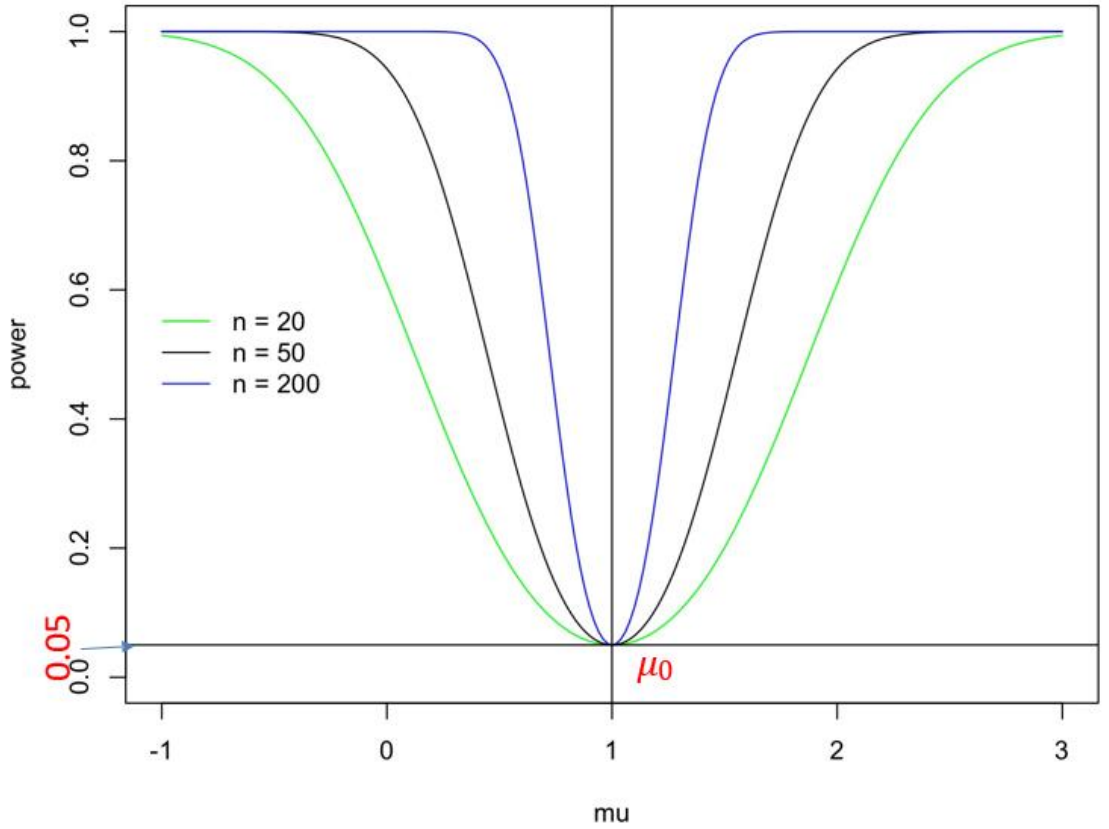
Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** $(1)\ H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0,$

$\mu_0 = 1, \quad \alpha = 0.05, \quad n = 50$



$\mu_0 = 1, \quad \alpha = 0.05, \quad \sigma^2 = 4$



Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

Discussion. Obviously, $\bar{X} = \sum_{i=1}^{n} X_i/n$ is a good estimate of μ, and H_0 is more likely false if $|\bar{X} - \mu_0|/S$ is large where S^2 is the sample variance. The rejection region can take the form of

$$D' = \{\boldsymbol{X} = (X_1, \cdots, X_n) : |\bar{X} - \mu_0|/S > A\}.$$

To determine the critical value A, we know that $\sqrt{n}(\bar{X} - \mu_0)/S \sim t_{n-1}$ under H_0. That is, when H_0 is true,

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t_{n-1}.$$

We can take T as test statistic, and the rejection region is

$$\{\boldsymbol{X} = (X_1, \cdots, X_n) : |T| > c\}.$$

Since the level of significance is α, it holds that

$$P(|T| > c \mid H_0) = P\left(\left|\frac{\sqrt{n}(\bar{X} - \mu_0)}{S}\right| > c \mid H_0\right) = \alpha,$$

therefore, $c = t_{n-1,\alpha/2}$, the upper $\alpha/2$ quantile of t_{n-1}. The rejection region is

$$D' = \left\{\boldsymbol{X} = (X_1, \cdots, X_n) : \left|\frac{\sqrt{n}(\bar{X} - \mu_0)}{S}\right| > t_{n-1,\alpha/2}\right\}$$

Question: D and D', which one is better?

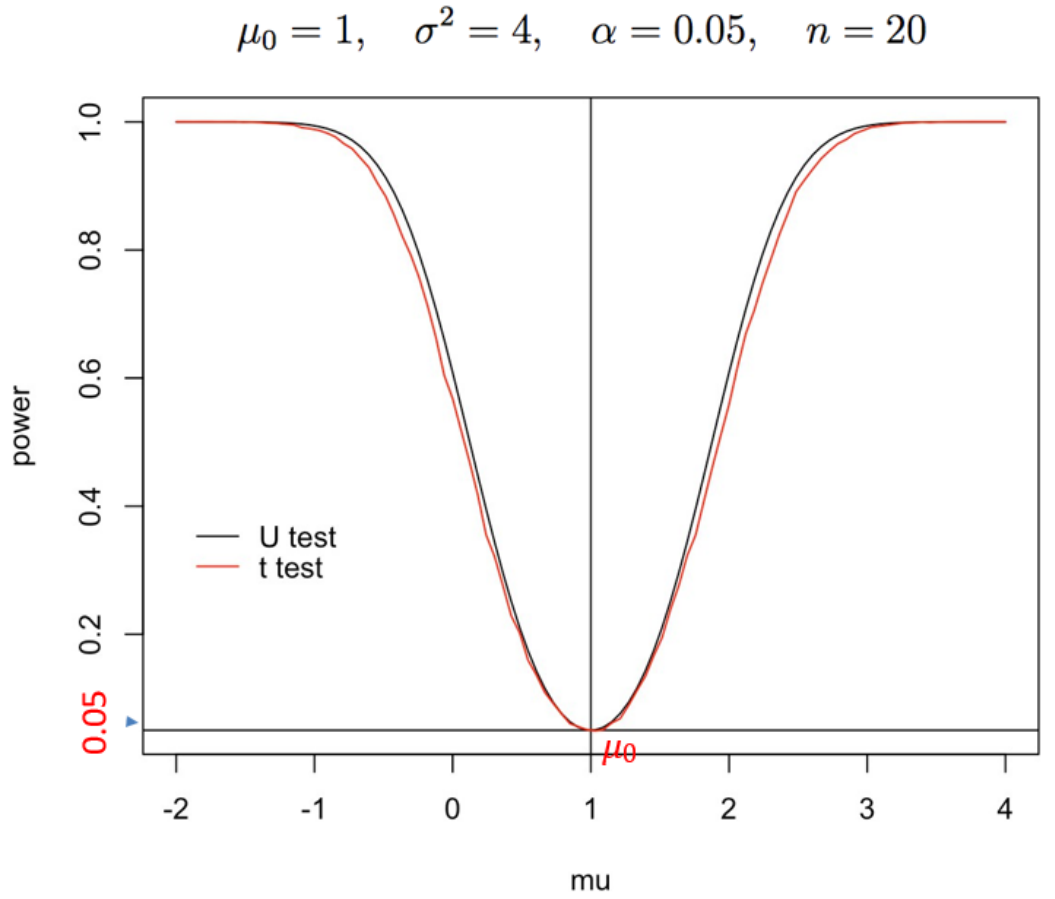
Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

Let $\Phi(\cdot)$ be the d.f. of $N(0,1)$, the power function of the test D' is

$$\pi_{\sigma^2}(\mu) = P_\mu\left(\left|\frac{\sqrt{n}(\bar{X}-\mu_0)}{S}\right| > t_{n-1,\alpha/2}\right)$$

It is complicated and depends on the value of S^2



Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (2) $H_0 : \mu \leq \mu_0 \longleftrightarrow H_1 : \mu > \mu_0,$

Discussion. Intuitively, the test statistic for problem (2) is still $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma$ and the rejection region is $\{\boldsymbol{X} = (X_1, \cdots, X_n) : \sqrt{n}(\bar{X} - \mu_0)/\sigma > c\}$. To determine c, let

$$P(Z > c \mid \mu = \mu_0) = P\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > c \mid \mu = \mu_0\right) = \alpha,$$

then $c = z_\alpha$. The rejection region

$$D_2^* = \left\{\boldsymbol{X} = (X_1, \cdots, X_n) : Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} > z_\alpha\right\}$$

is the desired one for test:

$$H_0^* : \mu = \mu_0 \longleftrightarrow H_1 : \mu > \mu_0$$

The question is: Whether D_2^* is the appropriate rejection region for test problem (2)? The main question is when $\mu < \mu_0$, whether the level of significance determined by the rejection region is still α?

$$\alpha(\theta) = \pi(\theta) = P_\theta(\text{rejecting } H_0) \leq \alpha, \quad \forall\, \theta \in \Theta_0$$

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (2) $H_0 : \mu \leq \mu_0 \longleftrightarrow H_1 : \mu > \mu_0$,

If the power function $\pi(\mu)$ (probability of type I error) is non-decreasing, then $\pi(\mu) \leq \pi(\mu_0) = P(Z > z_\alpha \mid \mu = \mu_0) = \alpha$, for any $\mu \leq \mu_0$.

Next, we show that $\pi(\mu)$ is non-decreasing. Let $\Phi(\cdot)$ be the d.f. of $N(0,1)$, then,

$$\begin{aligned}
\pi(\mu) &= P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} > z_\alpha\right) \\
&= P_\mu\left(\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} > z_\alpha + \frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right) \\
&= 1 - \Phi\left(z_\alpha + \frac{\sqrt{n}(\mu_0-\mu)}{\sigma}\right),
\end{aligned}$$

which is an increasing function. Therefore,

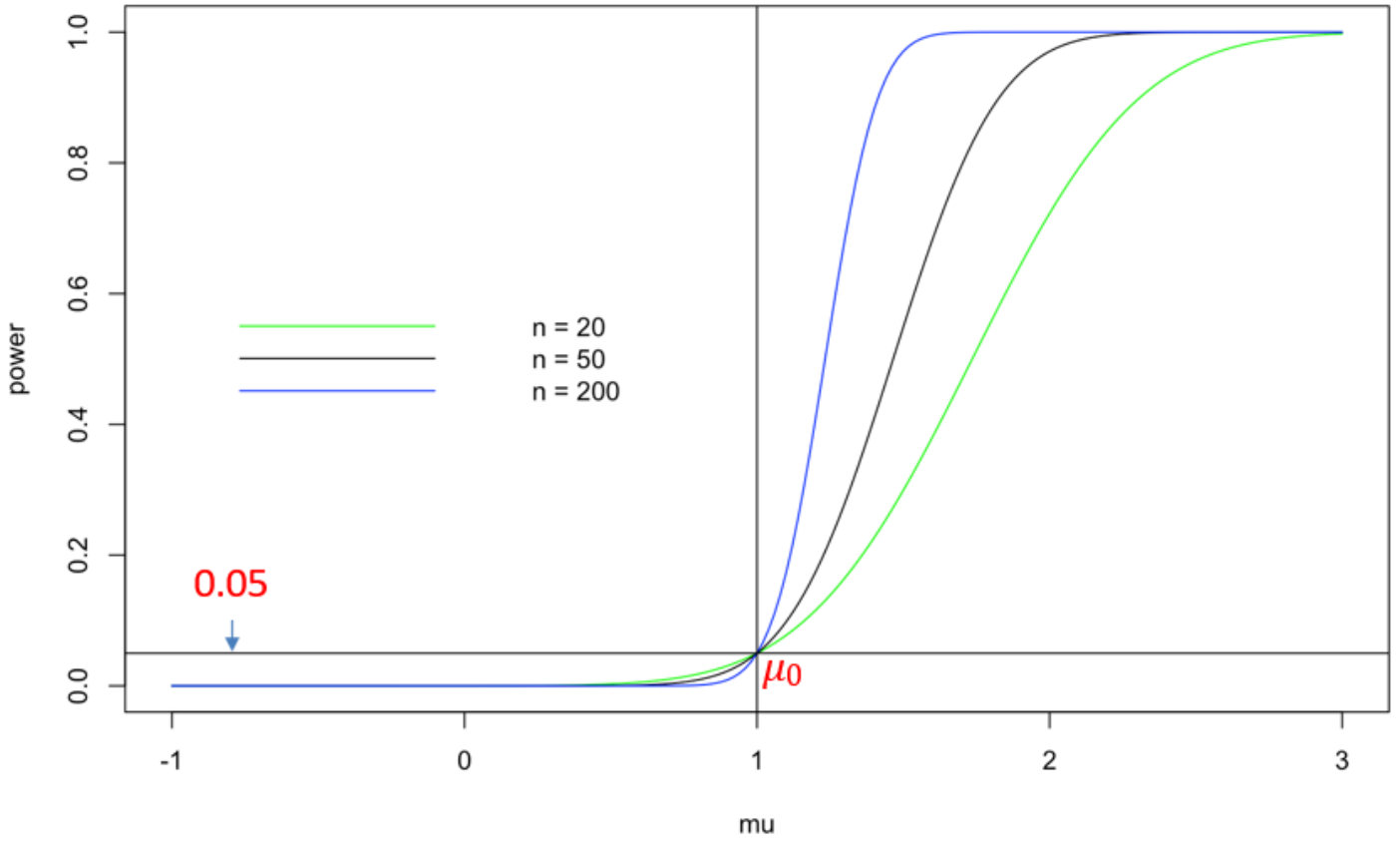
$$D_2 = D_2^* = \left\{\boldsymbol{X} = (X_1, \cdots, X_n) : Z = \frac{\sqrt{n}(\bar{X}-\mu_0)}{\sigma} > z_\alpha\right\}$$

is the rejection region of test problem (2) with level of significance α.

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 known** (2) $H_0 : \mu \leq \mu_0 \longleftrightarrow H_1 : \mu > \mu_0,$

$$\mu_0 = 1, \quad \alpha = 0.05, \quad \sigma^2 = 4$$



Testing Parameters in a Single Normal Distribution

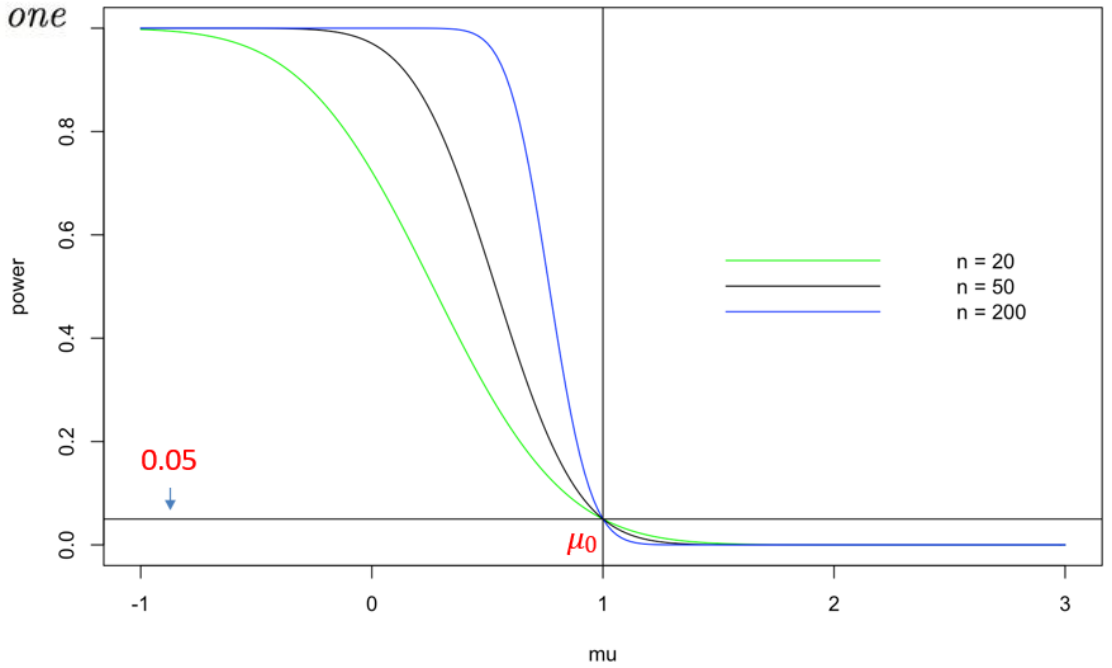
- **Test for μ when σ^2 known** (3) $H_0 : \mu \geq \mu_0 \longleftrightarrow H_1 : \mu < \mu_0.$

Discussion. By similar arguments as in test problem (2), the rejection region for test problem (3) with level of significance α is given by

$$D_3 = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} < -z_\alpha \right\}$$

The test method based on test statistic with $N(0,1)$ distribution is called *one sample U-test*

$\mu_0 = 1, \quad \alpha = 0.05, \quad \sigma^2 = 4$



Testing Parameters in a Single Normal Distribution

▶ Test for μ when σ^2 unknown

When σ^2 is unknown, we need to estimate it, for example, by the sample variance $S^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2/(n-1)$. And adopt the following test statistic

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \sim t_{n-1} \quad | \, \mu = \mu_0$$

By similar arguments as when σ^2 is known, we can obtain the rejection regions for test problem (1), (2), (3)

$$D_1 = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| > t_{n-1,\alpha/2} \right\}$$

$$D_2 = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > t_{n-1,\alpha} \right\}$$

$$D_3 = \left\{ \boldsymbol{X} = (X_1, \cdots, X_n) : \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} < -t_{n-1,\alpha} \right\}$$

respectively.

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 unknown** $\quad (1)\ H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0,$

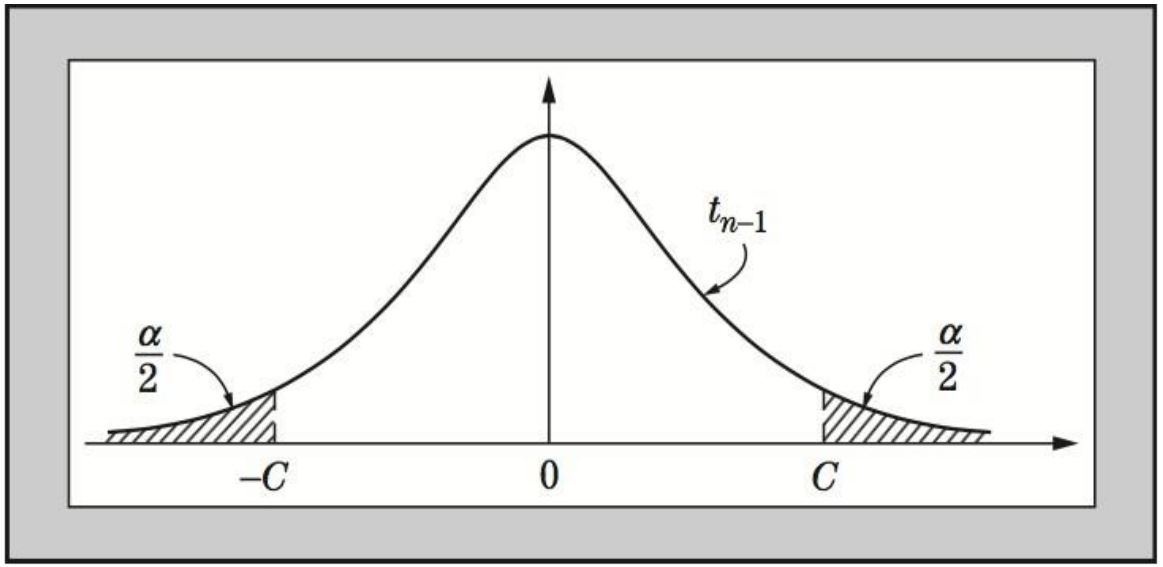


FIGURE 11.9

Rejection region of the hypothesis H_0 in (1) (the shaded areas), and the respective probabilities; Here $C = t_{n-1;\alpha/2}$.

Testing Parameters in a Single Normal Distribution

- **Test for μ when σ^2 unknown** (1) $H_0 : \mu = \mu_0 \longleftrightarrow H_1 : \mu \neq \mu_0$,

The power functions for tests (1), (2), (3) are

$$\pi_1(\mu) = P_\mu\left(\left|\frac{\sqrt{n}(\bar{X} - \mu_0)}{S}\right| > t_{n-1,\alpha/2}\right)$$

$$\pi_2(\mu) = P_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > t_{n-1,\alpha}\right)$$

$$\pi_3(\mu) = P_\mu\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{S} < -t_{n-1,\alpha}\right)$$

respectively. But these power functions do not have simple forms such as those when σ^2 is known. We can use Monte Carlo to approximate the power functions.

Testing Parameters in a Single Normal Distribution

Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from normal distribution $N(\mu, \sigma^2)$

	H_0	H_1	Test Statistic and its Distribution	Rejection Region
σ^2 known	$\mu = \mu_0$	$\mu \neq \mu_0$	$Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma$ $Z \mid \mu = \mu_0 \sim N(0,1)$	$\lvert Z \rvert > z_{\alpha/2}$
	$\mu \leq \mu_0$	$\mu > \mu_0$		$Z > z_{\alpha}$
	$\mu \geq \mu_0$	$\mu < \mu_0$		$Z < -z_{\alpha}$
σ^2 unknown	$\mu = \mu_0$	$\mu \neq \mu_0$	$T = \sqrt{n}(\bar{X} - \mu_0)/S$ $T \mid \mu = \mu_0 \sim t_{n-1}$	$\lvert T \rvert > t_{n-1,\alpha/2}$
	$\mu \leq \mu_0$	$\mu > \mu_0$		$T > t_{n-1,\alpha}$
	$\mu \geq \mu_0$	$\mu < \mu_0$		$T < -t_{n-1,\alpha}$

	H_0	H_1	Test Statistic and its Distribution	Rejection Region
μ known	$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$	$\chi^2_\mu = nS^2_\mu/\sigma_0^2$ $\chi^2_\mu \mid \sigma^2 = \sigma_0^2 \sim \chi^2_n$	$\frac{nS^2_\mu}{\sigma_0^2} < \chi^2_{n,1-\alpha/2}$ or $\frac{nS^2_\mu}{\sigma_0^2} > \chi^2_{n,\alpha/2}$
	$\sigma^2 \leq \sigma_0^2$	$\sigma^2 > \sigma_0^2$		$\frac{nS^2_\mu}{\sigma_0^2} > \chi^2_{n,\alpha}$
	$\sigma^2 \geq \sigma_0^2$	$\sigma^2 < \sigma_0^2$		$\frac{nS^2_\mu}{\sigma_0^2} < \chi^2_{n,1-\alpha}$
μ unknown	$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$	$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$ $\chi^2 \mid \sigma^2 = \sigma_0^2 \sim \chi^2_{n-1}$	$\frac{(n-1)S^2}{\sigma_0^2} < \chi^2_{n-1,1-\frac{\alpha}{2}}$ or $\frac{(n-1)S^2}{\sigma_0^2} > \chi^2_{n-1,\frac{\alpha}{2}}$
	$\sigma^2 \leq \sigma_0^2$	$\sigma^2 > \sigma_0^2$		$\frac{(n-1)S^2}{\sigma_0^2} > \chi^2_{n-1,\alpha}$
	$\sigma^2 \geq \sigma_0^2$	$\sigma^2 < \sigma_0^2$		$\frac{(n-1)S^2}{\sigma_0^2} < \chi^2_{n-1,1-\alpha}$

Hypothesis testing in two normal populations

Example 5. It is claimed that a new treatment is more effective than the standard treatment for prolonging the lives of terminal cancer patients. The standard treatment has been used for a long time, and from records in medical journals the mean survival period is known to have a certain numerical value (in years). The new treatment is administered to n patients and their duration of survival is recorded. On the basis of the existing journal information and the data gathered, check whether or not the claim made is supported.

Discussion. Suppose that the survival time for a terminal cancer patient treated with the standard treatment is a r.v. $X \sim N(\theta_1, \sigma_1^2)$. Likewise, let the r.v. Y stand for the survival time for such a patient subject to the new treatment, and let $Y \sim N(\theta_2, \sigma_2^2)$. Then the hypothesis to be tested here is

$$H_0 : \theta_2 = \theta_1 \longleftrightarrow H_1 : \theta_2 > \theta_1.$$

Testing Parameters in a Single Normal Distribution

▶ Test for $\mu_1 - \mu_2$ in two Normal Population

Let $\boldsymbol{X} = (X_1, \cdots, X_m)$ be a random sample from normal distribution $N(\mu_1, \sigma_1^2)$, Let $\boldsymbol{Y} = (Y_1, \cdots, Y_n)$ be a random sample from $N(\mu_2, \sigma_2^2)$. Suppose that X_i's and Y_i's are independent, consider the following hypothesis testing problems:

(7) $H_0 : \mu_1 - \mu_2 = \mu_0 \longleftrightarrow H_1 : \mu_1 - \mu_2 \neq \mu_0$,

(8) $H_0 : \mu_1 - \mu_2 \leq \mu_0 \longleftrightarrow H_1 : \mu_1 - \mu_2 > \mu_0$,

(9) $H_0 : \mu_1 - \mu_2 \geq \mu_0 \longleftrightarrow H_1 : \mu_1 - \mu_2 < \mu_0$.

where μ_0 and the level of significance α is preassigned (or given).

- σ_1^2 and σ_2^2 are known
- $\sigma_1^2 = \sigma_2^2 = \sigma^2$ is unknown
- σ_1^2, σ_2^2 are unknown
- $m = n$ (X_i's and Y_i's may not be independent)

Testing Parameters in a Single Normal Distribution

- **Test for $\mu_1 - \mu_2$ when σ_1^2, σ_2^2 known**

Discussion. Consider two-side test problem (7),

$$H_0 : \mu_1 - \mu_2 = \mu_0 \longleftrightarrow H_1 : \mu_1 - \mu_2 \neq \mu_0$$

Obviously, $\bar{X} - \bar{Y}$ is a good estimate of $\mu_1 - \mu_2$, and H_0 is more likely false if $|\bar{X} - \bar{Y} - \mu_0|$ is large. The rejection region can take the form of $D = \{(X_1, \cdots, X_m; Y_1, \cdots, Y_n) : |\bar{X} - \bar{Y} - \mu_0| > A\}$. To determine the critical value A, we know that when σ_1^2, σ_2^2 are known and when H_0 is true,

$$U = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}} \sim N(0, 1).$$

We take U as test statistic, and the rejection region is

$$D_7 = \{(X_1, \cdots, X_m; Y_1, \cdots, Y_n) : |U| > c\}.$$

To determine c, let $\theta = \mu_2 - \mu_1$ and let

$$\alpha = P_\theta(|U| > c \mid H_0) = P\left(\left|\frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}}\right| > c \mid H_0\right),$$

therefore, $c = z_{\alpha/2}$. The rejection region is

Testing Parameters in a Single Normal Distribution

- **Test for $\mu_1 - \mu_2$ when σ_1^2, σ_2^2 known**

therefore, $c = z_{\alpha/2}$. The rejection region is

$$D = \left\{ (X_1, \cdots, X_m; Y_1, \cdots, Y_n) : \left| \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}} \right| > z_{\alpha/2} \right\}$$

Similar to test problems (2) and (3), we can obtain the rejection regions for test problems (8), (9) with level of significance α are

$$D_8 = \left\{ (X_1, \cdots, X_m; Y_1, \cdots, Y_n) : U = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}} > z_{\alpha} \right\}$$

$$D_9 = \left\{ (X_1, \cdots, X_m; Y_1, \cdots, Y_n) : U = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}} < -z_{\alpha} \right\}$$

respectively.

The above tests are called ***two sample U test***.

Testing Parameters in a Single Normal Distribution

- **Test for $\mu_1 - \mu_2$ when $\sigma_1^2 = \sigma_2^2 = \sigma^2$ unknown**

Discussion. When $\sigma_1^2 = \sigma_2^2 = \sigma^2$ is unknown, we can use the following test statistic with t_{m+n-2} distribution under H_0:

$$T = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}\left(\frac{1}{m} + \frac{1}{n}\right)}} \sim t_{m+n-2} \quad | \ \mu_1 - \mu_2 = \mu_0$$

The tests are called ***two sample t test.***

	H_0	H_1	Test Statistic and its Distribution	Rejection Region
σ_1^2, σ_2^2 known	$\mu_1 - \mu_2 = \mu_0$	$\mu_1 - \mu_2 \neq \mu_0$	$U = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\sigma_1^2/m + \sigma_2^2/n}}$ $U \mid \mu_1 - \mu_2 = \mu_0 \sim N(0,1)$	$\lvert U \rvert > z_{\alpha/2}$
	$\mu_1 - \mu_2 \leq \mu_0$	$\mu_1 - \mu_2 > \mu_0$		$U > z_\alpha$
	$\mu_1 - \mu_2 \geq \mu_0$	$\mu_1 - \mu_2 < \mu_0$		$U < -z_\alpha$
$\sigma_1^2 = \sigma_2^2 = \sigma^2$ unknown	$\mu_1 - \mu_2 = \mu_0$	$\mu_1 - \mu_2 \neq \mu_0$	$T = \frac{\bar{X} - \bar{Y} - \mu_0}{\sqrt{\frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}\left(\frac{1}{m} + \frac{1}{n}\right)}}$ $T \mid \mu_1 - \mu_2 = \mu_0 \sim t_{m+n-2}$	$\lvert T \rvert > t_{m+n-2,\alpha/2}$
	$\mu_1 - \mu_2 \leq \mu_0$	$\mu_1 - \mu_2 > \mu_0$		$T > t_{m+n-2,\alpha}$
	$\mu_1 - \mu_2 \geq \mu_0$	$\mu_1 - \mu_2 < \mu_0$		$T < -t_{m+n-2,\alpha}$

Three Classical Asymptotic Tests for MLE

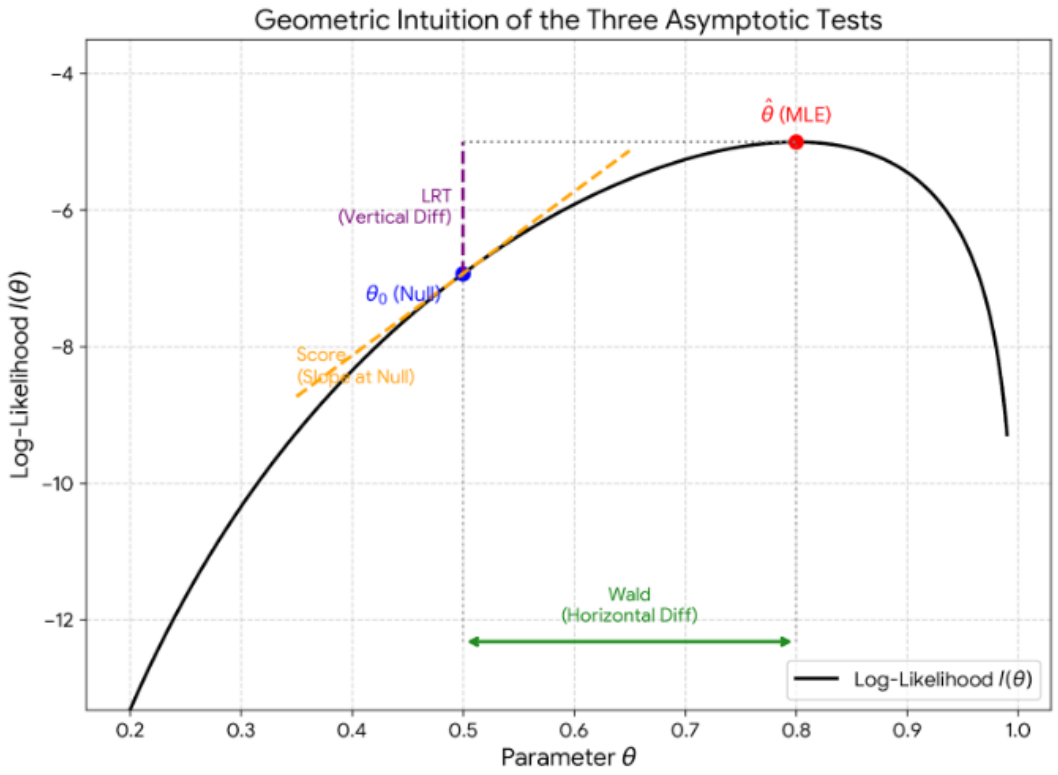
In large samples, there are three asymptotically equivalent tests for the null hypothesis $H_0: \theta = \theta_0$
All three test statistics converge to a χ_1^2 distribution (or standard Normal) in 1-dimensional case.

- **Likelihood Ratio Test (LRT):**

  Evaluates the vertical difference in the log-likelihood function.

- **Wald Test:** Evaluates the horizontal distance between the estimate $\hat{\theta}$ and θ_0.

- **Score Test:** Evaluates the gradient (slope) of the log-likelihood function at θ_0.



Likelihood Ratio Test (LRT)

Compares the maximum log-likelihood under the full model $l(\hat{\theta})$ to the maximum log-likelihood under the restricted null model $l(\theta_0)$.

- **Test Statistic**: $W = 2[l(\hat{\theta}) - l(\theta_0)] \sim \chi_1^2]$.
- **Estimation**: Requires fitting the model twice (both under H_1 to get $\hat{\theta}$ and under H_0).
- **Properties**: Highly reliable in small samples. Invariant to reparameterization (e.g., testing θ vs $\log\theta$ give the same result).

Wald Test

Measures how far the estimated parameter MLE $\hat{\theta}$ is from the hypothesized value θ_0, relative to its standard error.

- **Test Statistic(1D)**: $Z_W = \frac{\hat{\theta}-\theta_0}{\sqrt{I^{-1}(\hat{\theta})}} \sim N(0,1) \Rightarrow W_p = Z_W^2 = (\hat{\theta}-\theta_0)^2 I(\hat{\theta}) \sim \chi_1^2]$.
- **Estimation**: Requires calculating the unconstrained MLE $\hat{\theta}$ and estimationg its variance $I(\hat{\theta})$ or observed information $J(\hat{\theta})$.
- **Properties**: Computationally easy (only fit full model). However, it is not invariant to reparameterization and can be inaccurate for small n.

Score Test

Evaluates the score function $U(\theta) = l'(\theta)$ at the null value θ_0. If H_0 is true, the slope should be near zero.

- **Distribution of Score**: $U(\theta) \sim N\big(0, I(\theta)\big)$.
- **Test Statistic(1D)**: $Z_S = \frac{l'(\theta_0)}{\sqrt{I(\theta_0)}} \sim N(0,1) \Rightarrow W_u = Z_S^2 = \frac{[l'(\theta_0)]^2}{I(\theta_0)} \sim \chi_1^2]$.
- **Estimation**: Does not require computing the MLE $\hat{\theta}$. Only requires evaluating the model under H_0.
- **Properties**: Useful when fitting the full model is computationally expensive.

Three Classical Asymptotic Tests for MLE


```r
# Simulate biological data: binary outcome (e.g., Disease Yes/No)
set.seed(123)
x <- rnorm(100) # Biomarker level
y <- rbinom(100, 1, prob = plogis(0.5 + 0.8 * x))

# Fit Models
fit_full <- glm(y ~ x, family = binomial)
fit_null <- glm(y ~ 1, family = binomial)

# 1. Wald Test (Provided by default in summary)
summary(fit_full)$coefficients
# The 'z value' and 'Pr(>|z|)' columns test H0: beta = 0 using Wald

# 2. Likelihood Ratio Test (LRT)
anova(fit_null, fit_full, test = "LRT")
# Alternatively: 2 * as.numeric(logLik(fit_full) - logLik(fit_null))

# 3. Score Test (Rao's Test)
anova(fit_null, fit_full, test = "Rao")
```

```
> # 1. Wald Test (Provided by default in summary)
> summary(fit_full)$coefficients
             Estimate Std. Error  z value     Pr(>|z|)
(Intercept) 0.636986  0.2275513 2.799307 0.0051212416
x           0.976607  0.2891357 3.377677 0.0007310079

> # 2. Likelihood Ratio Test (LRT)
> anova(fit_null, fit_full, test = "LRT")
Analysis of Deviance Table

Model 1: y ~ 1
Model 2: y ~ x
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        99     129.49
2        98     115.30  1   14.185 0.0001656 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


> # 3. Score Test (Rao's Test)
> anova(fit_null, fit_full, test = "Rao")
Analysis of Deviance Table

Model 1: y ~ 1
Model 2: y ~ x
  Resid. Df Resid. Dev Df Deviance   Rao  Pr(>Chi)
1        99     129.49
2        98     115.30  1   14.185 13.15 0.0002874 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Hypothesis Testing in linear regression model

Linear Regression Model. Denote $\beta = (\beta_0, \beta_1, \cdots \beta_p)$, $x_i = (1, x_{i1}, \cdots, x_{ip})'$

$$y_i = x_i^T \beta + \epsilon_i, \quad i = 1, \cdots, n$$

Matrix form

$$Y = X\beta + \epsilon$$

- $Y = (y_1, \cdots, y_n)'$: n-dim vector of response (random)
- $X = (x_1', \cdots, x_n')'$: $n \times (p+1)$ design matrix (deterministic/fixed)
- $\epsilon = (\epsilon_1, \cdots \epsilon_n)'$: n-dim vector of random error (random)

Normal assumption:

- $\epsilon_i,\ i.i.d. \sim N(0, \sigma^2)$
- The p.d.f. of y_i is

$$f(\boldsymbol{y} \mid \boldsymbol{x}, \beta, \sigma^2) = \prod_{i=1}^{n} f(y_i \mid x_i, \beta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y_i - x_i^T\beta)^2}{2\sigma^2}\right\}$$

$$\boldsymbol{y} = (y_1, \cdots, y_n)' \text{ and } \boldsymbol{x} = (x_1', \cdots, x_n')'$$

- Point estimate (OLS and MLE)

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(\hat{\epsilon}_i^2) = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - x_i^T\hat{\beta})^2 = \frac{RSS}{n-p-1} \text{ (bias-corrected)}$$

$$RSS = \sum_{i=1}^{n}(y_i - x_i^T\hat{\beta})^2 = \|Y - X\hat{\beta}\|_2^2$$

Hypothesis Testing in linear regression model

The ordinary least squares (OLS) estimator minimizes the residual sum of squares

$$\hat{\beta} \in \arg\min_{\beta \in \mathbb{R}^p} S(\beta), \qquad S(\beta) := \|y - X\beta\|_2^2 = (y - X\beta)^\top (y - X\beta).$$

Expanding the quadratic form yields

$$S(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

Taking gradient with respect to β gives

$$\nabla_\beta S(\beta) = -2X^\top y + 2X^\top X\beta.$$

At the minimizer, the first-order condition (normal equations) is

$$\nabla_\beta S(\hat{\beta}) = 0 \quad \Longleftrightarrow \quad X^\top X\hat{\beta} = X^\top y.$$

If $X^\top X$ is invertible (e.g., X has full column rank), then the OLS estimator has the closed form

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

Hypothesis Testing in linear regression model

Let $\hat{\beta} = (X^\top X)^{-1} X^\top y$ denote the OLS estimator. The residual vector is

$$\hat{\varepsilon} = y - X\hat{\beta}.$$

The residual sum of squares (RSS) is

$$\text{RSS} = \|\hat{\varepsilon}\|_2^2 = (y - X\hat{\beta})^\top (y - X\hat{\beta}).$$

The unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n - p - 1} (y - X\hat{\beta})^\top (y - X\hat{\beta}).$$

Hypothesis Testing in linear regression model

Theorem. In the linear regression model

$$Y = X\beta + \varepsilon,$$

suppose that the design matrix is deterministic (not random) and the random errors

$$\varepsilon_1, \dots, \varepsilon_n \, i.i.d. \sim N(0, \sigma^2),$$

let $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ be the MLE estimator of the regression coefficient $\beta = (\beta_0, \beta_1, \dots, \beta_p)$ and $\hat{\sigma}^2$ be the bias-corrected variance estimator, then the following holds:

(1) $E\hat{\beta} = \beta$. In particular, $E\hat{\beta}_j = \beta_j$, $j = 0, 1, \cdots, p$.

(2) $\hat{\beta} \sim N(\beta, \sigma^2(X^TX)^{-1})$. In particular, $\hat{\beta}_j \sim N\left(\beta_j, \sigma^2\left[(X^TX)^{-1}\right]_{j+1,j+1}\right)$.

(3) $\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p-1}$.

(4) $\hat{\beta}$ is independent of $\hat{\sigma}^2$.

Hypothesis Testing in linear regression model

- **Confidence Interval**

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{j+1,j+1}} \sim t_{n-p-1}$$

Therefore, a $1-\alpha$ confidence interval for β_j is

$$\left[\hat{\beta}_j - \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{j+1,j+1} t_{n-p-1,\alpha/2},\ \hat{\beta}_j + \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{j+1,j+1} t_{n-p-1,\alpha/2}\right]$$

A $1-\alpha$ confidence interval for σ^2 is

$$\left[\frac{(n-p-1)\hat{\sigma}^2}{\chi^2_{n-p-1,\alpha/2}},\ \frac{(n-p-1)\hat{\sigma}^2}{\chi^2_{n-p-1,1-\alpha/2}}\right]$$

A $1-\alpha$ confidence region for (β_1, β_2) is

$$\left[\hat{\beta}_1 - \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{2,2} t_{n-p-1,\alpha/4},\ \hat{\beta}_1 + \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{2,2} t_{n-p-1,\alpha/4}\right]$$
$$\times \left[\hat{\beta}_2 - \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{3,3} t_{n-p-1,\alpha/4},\ \hat{\beta}_2 + \hat{\sigma}\left[(X^TX)^{-1}\right]^{1/2}_{3,3} t_{n-p-1,\alpha/4}\right]$$

Hypothesis Testing in linear regression model

▶ *t*-test in Linear Regression

Discussion. Test the hypothesis: (at the level of significance α)

$$H_{0j}: \beta_j = 0 \longleftrightarrow H_1: \beta_j \neq 0, \quad j = 0, 1, \cdots, p.$$

Obviously, $\hat{\beta}_j$ is a good estimate of β_j and the rejection region could take the form

$$D_j = \left\{ \{(y_i, x_i), i = 1, \cdots, n\} : \frac{|\hat{\beta}_j|}{\hat{\sigma}\left[(X^T X)^{-1}\right]_{j+1,j+1}^{1/2}} > c_j \right\}$$

We need to determine the critical value c_j. By Theorem 1 and under H_0

$$T_j = \frac{\hat{\beta}_j}{\hat{\sigma}\left[(X^T X)^{-1}\right]_{j+1,j+1}^{1/2}} \sim t_{n-p-1}$$

By the level of significance α, we can obtain the critical value $c_j = t_{n-p-1,\alpha/2}$. Therefore, the test with level of significance α is

$$\varphi(X, Y) = \begin{cases} 1, & \frac{|\hat{\beta}_j|}{\hat{\sigma}[(X^T X)^{-1}]_{j+1,j+1}^{1/2}} > t_{n-p-1,\alpha/2} \\ 0, & \text{otherwise} \end{cases}$$

Let t_j be the observed value of T_j, the p-value of the test is

$$p = P_{\beta_j=0}\left(|T_j| > |t_j|\right) = P\left(|t_{n-p-1}| > |t_j|\right)$$

Hypothesis Testing and P-Value

Hypothesis Testing

In regression $Y = \beta_0 + \beta_1 X_1 + \epsilon$, we test:

- **Null (H_0):** $\beta_1 = 0$ (No effect)
- **Alternative (H_1):** $\beta_1 \neq 0$ (Effect exists)

Let T be a test statistic, then the p-value is:

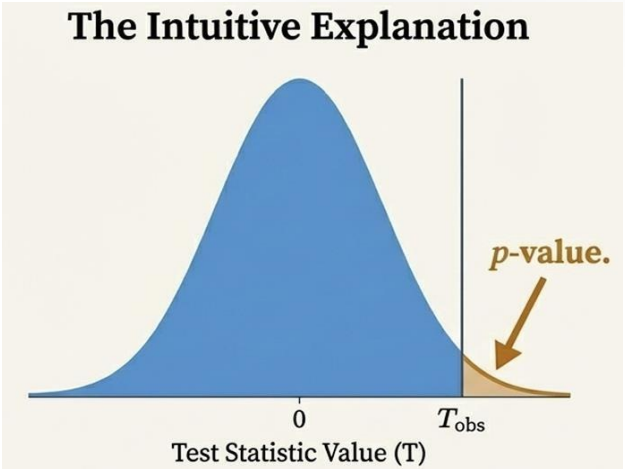
$$p = P(T \geq T_{obs} | H_0)$$

where T_{obs} is the observed value of the test statistic.

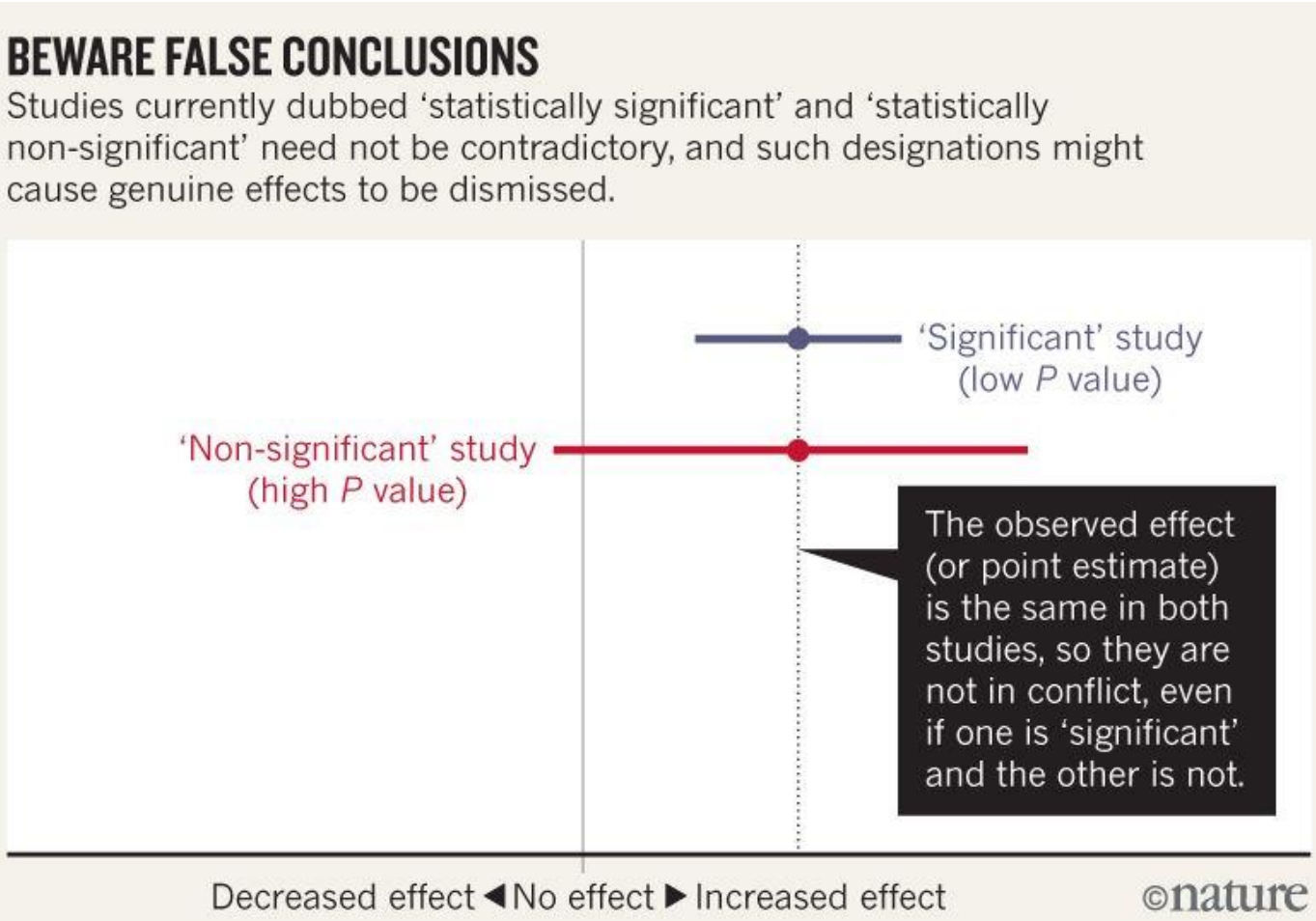
The probability of observing a result at least as extreme as the data under

```
             Estimate   Std. Error  t value   Pr(>|t|)
(Intercept)   5.79778    2.76324    2.098    0.0385 *
x             0.47376    0.05344    8.865   3.5e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1
```

It quantifies the evidence against the null hypothesis.

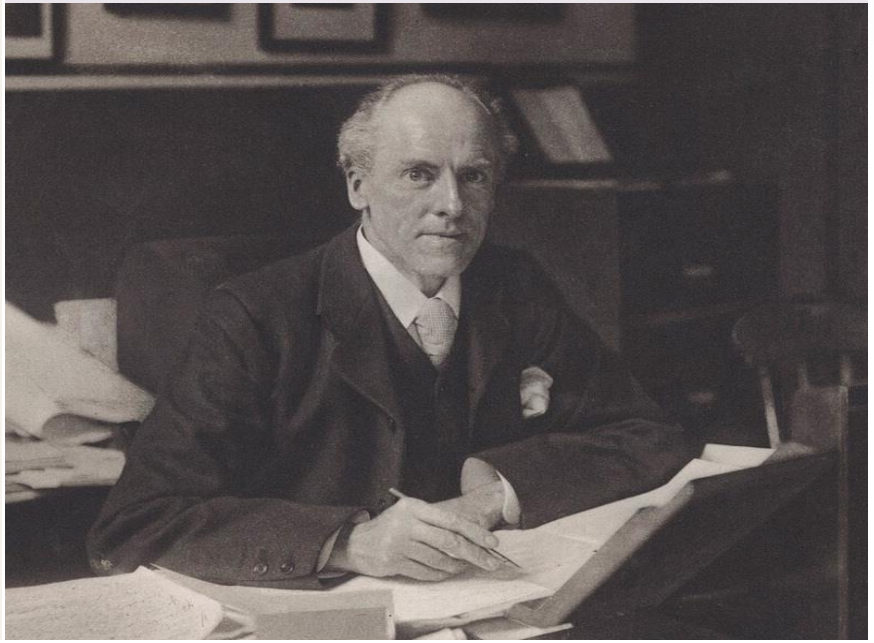


P-Value



From my point of view, the tests are used to ascertain whether a reasonable graduation curve has been achieved, not to assert whe ther one or another hypothesis is true or false.

Karl Pearson



What are P-Values?

Occupational Exposure to Extremely Low Frequency Magnetic Fields and Mortality from Cardiovascular Disease FREE

Niclas Håkansson, Per Gustavsson, Antonio Sastre, Birgitta Floderus

American Journal of Epidemiology, Volume 158, Issue 6, 15 September 2003, Pages 534–542, https://doi.org/10.1093/aje/kwg197

Published: 15 September 2003

The authors found a low-level increase in AMI risk in the highest exposure group (relative risk = **1.3**, 95% confidence interval: **0.9**, **1.9**) and suggestions of an exposure-response relation (**p = 0.02**).

What does it mean by “p=0.02”?

A. The probability that the exposure-response relationship is due to the chance alone is 0.02.

B. The probability that the null hypothesis (i.e., there is no exposure-response relationship) is false is 0.02.

C. The probability that the investigators make type I error if they conclude that the association is real is 2%.

D. It doesn’t really matter because it is just a scientific convention that if $p<0.05$, then the association is significant.

E. None of above.

What are P-Values?

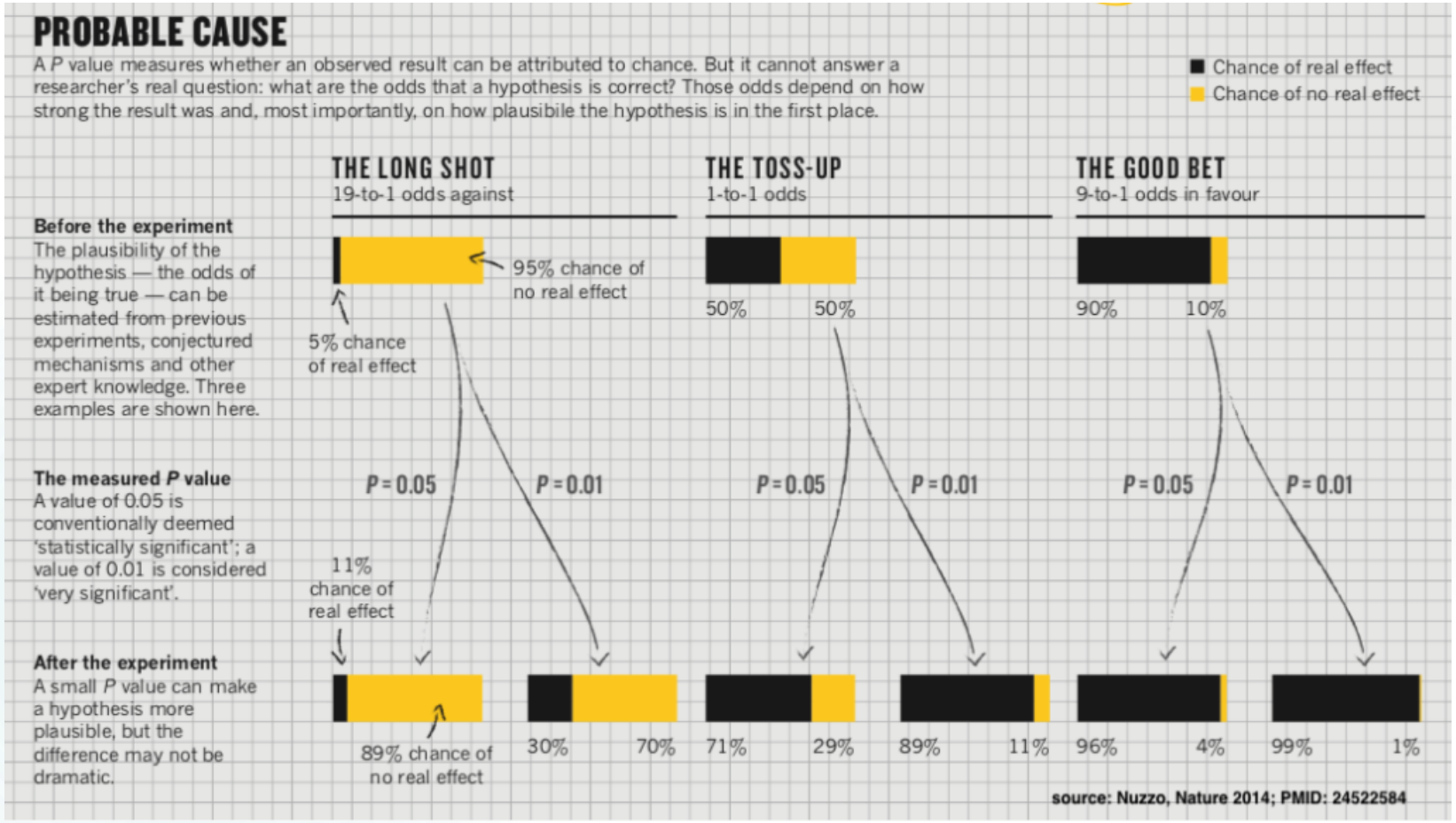
MYTH:

"A p-value of 0.02 means there is a 98% chance that the hypothesis is true."

The authors found a low-level increase in AMI risk in the highest exposure group (relative risk = **1.3**, 95% confidence interval: **0.9**, **1.9**) and suggestions of an exposure-response relation (**p = 0.02**).

If we did a similar study design again, the probability that we would obtain a similar or greater level of association than found in these data, if the null hypothesis (of no exposure-response relationship) is true, is 0.02.

Widespread Misunderstanding



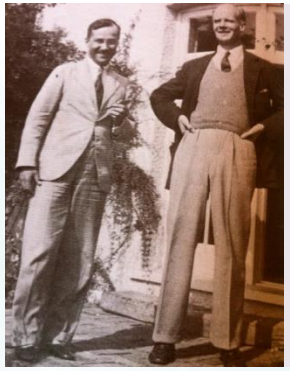
R. NUZZO; SOURCE: T. SELLKE ET AL. AM. STAT. 55, 62–71 (2001)

Interpretation of p-value



Probability that we would obtain a more extreme result than the actual one if the null hypothesis is true.

—— *Fisher*



No test based upon a theory of probability can by itself provide any valuable evidence of the truth or falsehood of a hypothesis.

—— *Neyman and Pearson*

No magic of *"P<0.05"*

Over interpretation in both directions "significant" and "non-significant".

Code Example: Calculating and Interpreting P-values

Here is a standard workflow for calculating and using a p-value in R,
using a Two-Sample T-test to compare gene expression levels between two groups.

```r
# 1. Simulate data: Gene expression levels (e.g., n=30 per group)
set.seed(123)
healthy_group <- rnorm(30, mean = 15.0, sd = 2.0)
disease_group <- rnorm(30, mean = 16.5, sd = 2.0)

# 2. Perform a two-sample t-test (H0: Means are equal)
t_test_result <- t.test(disease_group, healthy_group)

# 3. View the full statistical summary
print(t_test_result)
# The output will show: t = 3.28, df = 57.5, p-value = 0.0017
# It also provides the 95% Confidence Interval.

# 4. Extract and use the p-value programmatically
p_val <- t_test_result$p.value
alpha <- 0.05  # Set significance level

if(p_val < alpha) {
  cat("p-value (", p_val, ") < 0.05. \nConclusion: Reject H0. Significant difference observed.\n")
} else {
  cat("p-value (", p_val, ") >= 0.05. \nConclusion: Fail to reject H0.\n")
}
```

```
> # 3. View the full statistical summary
> print(t_test_result)

        Welch Two Sample t-test

data:  disease_group and healthy_group
t = 4.1469, df = 56.559, p-value = 0.0001142
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.008683 2.893085
sample estimates:
mean of x mean of y
 16.85668  14.90579


p-value ( 0.0001142107 ) < 0.05.
Conclusion: Reject H0. Significant difference observed.
```

Code Example: Testing in Linear Regression

In a simple linear regression model $Y = \beta + \beta_1 X + \varepsilon$, our primary interest is often testing the slope parameter: H_0: $\beta_1 = 0$.

By default, R utilizes a t-test (which is a 1-dimensional Wald test) to compute the p-value for each coefficient.

```r
# 1. 模拟临床数据：50名患者的 BMI 与 收缩压 (SBP)
set.seed(42)
n_patients <- 50
BMI <- rnorm(n_patients, mean = 24, sd = 3.5)

# 设定真实的线性关系：SBP = 80 + 1.5 * BMI + 随机噪声
SBP <- 80 + 1.5 * BMI + rnorm(n_patients, mean = 0, sd = 8)

# 2. 拟合普通最小二乘法 (OLS) 线性模型
fit_lm <- lm(SBP ~ BMI)

# 3. 输出完整的统计检验摘要
summary(fit_lm)
```

```r
# 4. 提取特定的检验结果用于报告
# 提取 BMI 斜率的 t 统计量和 p-value
t_stat <- summary(fit_lm)$coefficients["BMI", "t value"]
p_val  <- summary(fit_lm)$coefficients["BMI", "Pr(>|t|)"]

cat("检验 H0: beta_1 = 0 \n")
cat("t-statistic:", round(t_stat, 3), "\n")
cat("p-value:", format.pval(p_val, digits = 3), "\n")

# 5. 计算回归系数的 95% 置信区间
confint(fit_lm, level = 0.95)
```

```
> # 3. 输出完整的统计检验摘要
> summary(fit_lm)

Call:
lm(formula = SBP ~ BMI)

Residuals:
    Min      1Q  Median      3Q     Max
-21.595  -4.888   1.466   4.811  12.413

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  91.2843     6.2292  14.654  < 2e-16 ***
BMI           1.0611     0.2573   4.123 0.000147 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.26 on 48 degrees of freedom
Multiple R-squared:  0.2616,    Adjusted R-squared:  0.2462
F-statistic:    17 on 1 and 48 DF,  p-value: 0.0001472
```

```
> # 4. 提取特定的检验结果用于报告
> # 提取 BMI 斜率的 t 统计量和 p-value
> t_stat <- summary(fit_lm)$coefficients["BMI", "t value"]
> p_val  <- summary(fit_lm)$coefficients["BMI", "Pr(>|t|)"]
>
> cat("检验 H0: beta_1 = 0 \n")
检验 H0: beta_1 = 0
> cat("t-statistic:", round(t_stat, 3), "\n")
t-statistic: 4.123
> cat("p-value:", format.pval(p_val, digits = 3), "\n")
p-value: 0.000147
>
> # 5. 计算回归系数的 95% 置信区间
> confint(fit_lm, level = 0.95)
                2.5 %     97.5 %
(Intercept) 78.759689 103.808898
BMI          0.543692   1.578518
```

Statistical vs. Scientific Significance

Statistical Significance ≠ Scientific Significance

Statistical Significance (The P-value)

- Tells us about the *reliability* of the evidence against H_0.
- Answers: "Is this effect likely due to chance?"
- **Strongly depends on sample size (n).** With $n \to \infty$, even biologically trivial effects yield $p < 0.001$.

Scientific Significance (Effect Size)

- Tells us about the *magnitude* and *practical importance* of the effect.
- Answers: "Does this effect actually matter in the real world?"
- **Independent of sample size.**

Example in Biostatistics: A new weight-loss drug reduces weight by exactly 0.01 kg compared to placebo. In a massive trial of 1,000,000 patients, this difference might have $p < 10^{-6}$ (Highly Statistically Significant), but it is completely meaningless for patient health (Not Clinically Significant). Always report **Effect Sizes** and **Confidence Intervals** alongside p-values!

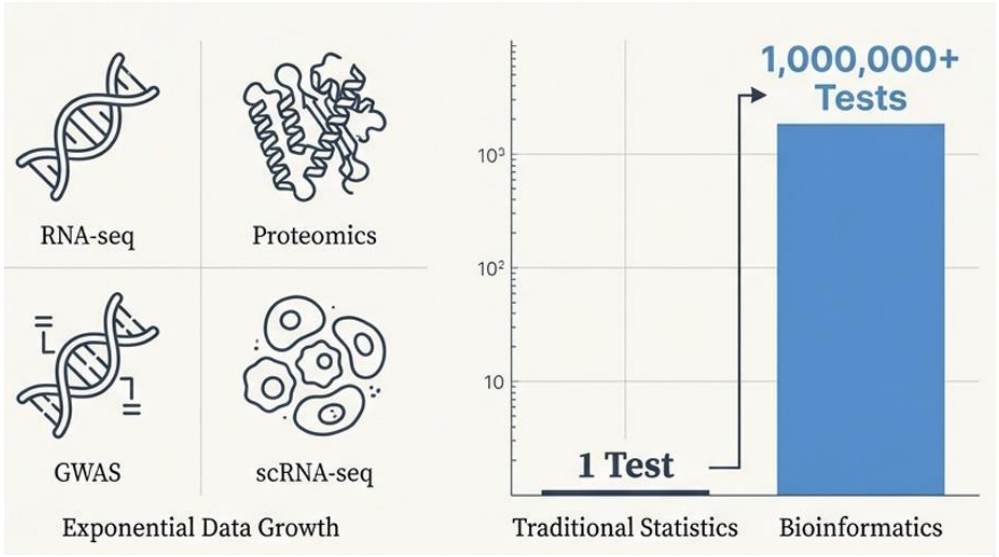
Small Samples vs. Large Samples

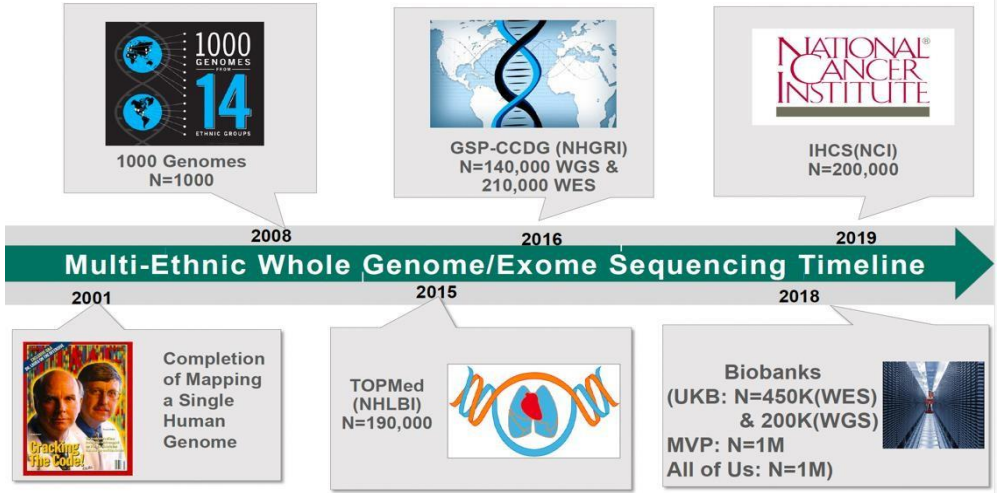
Sample Instability

- Small samples have low power.
- They often produce "over-estimated" effect sizes.
- Small samples are prone to publication bias: Only "extreme" small studies survive to journals.

The Trap of Large Samples

- High-throughput data generates thousands/millions of simultaneous hypotheses.
- How to define the significance & find interesting discovery?
- Reproducibility? Reliable biological insights.





Null Hypothesis and Interpretation

Wilcoxon test

- Non-parametric: Compare two distribution difference
- Mean difference / the dispersion difference

GLM test

- Parametric: coefficient test
- Model the mean difference

Classical Interpretation

α	0.1	0.05	0.025	0.01
Strength of Evidence	Borderline	Moderate	Substantial	Strong

- With big sample sizes, it is easy to have small p-value even when the estimated effect is small.
- Performing many tests can lead to many false discoveries.
- It does not measure reproducibility.

Multiple Testing

Hypotheses

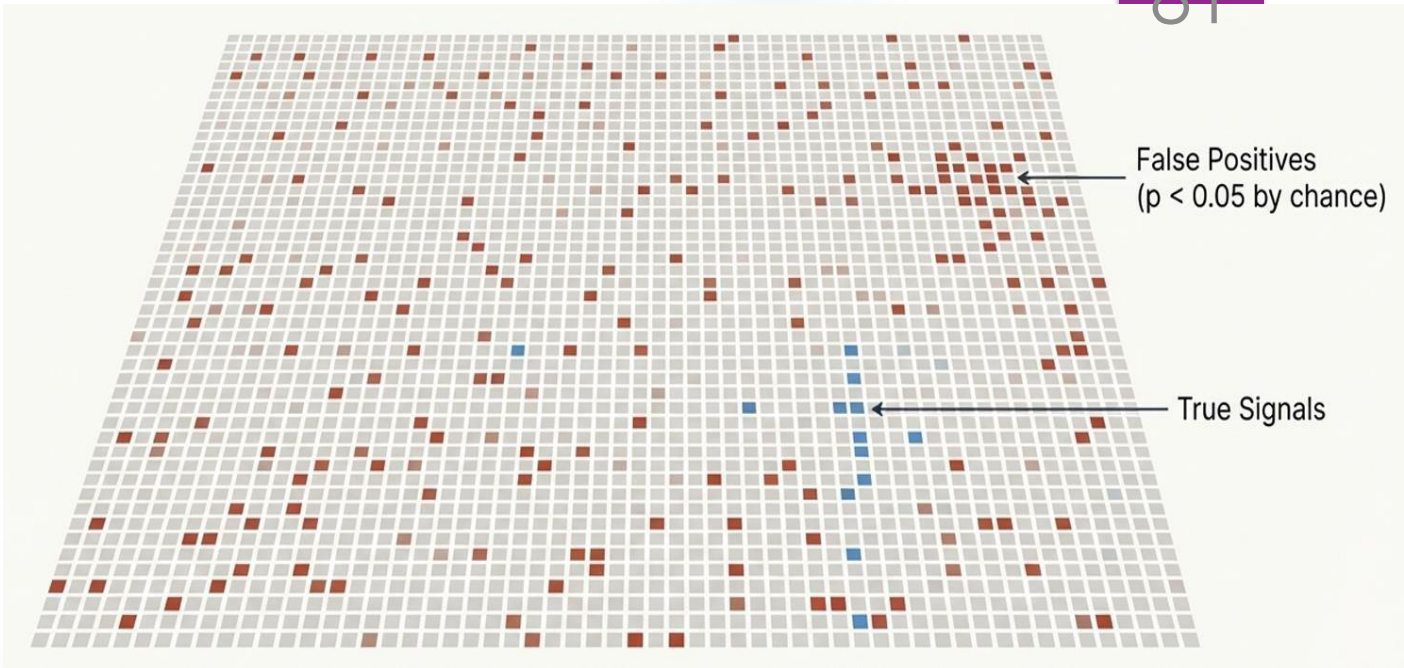
$H_{0,i}$: Gene i is NOT differentially expressed.

$H_{1,i}$: Gene i is differentially expressed.

The Challenge

If we use a standard threshold of $\alpha = 0.05$ for 20,000

independent tests...



1,000

False Positives Expected

$20{,}000 \times 0.05 = 1{,}000$

(Purely by chance!)

The Reproducibility Crisis

> *"Why most published research findings are false." (Ioannidis, 2005)*

Many top findings in biomedicine and psychology cannot be reproduced in independent labs.

Causes: Pressure to publish "positive" results, selective reporting, and underpowered studies.

- Low power.
- Publish-or-perish pressure.
- Absence of "Null results" in journals.

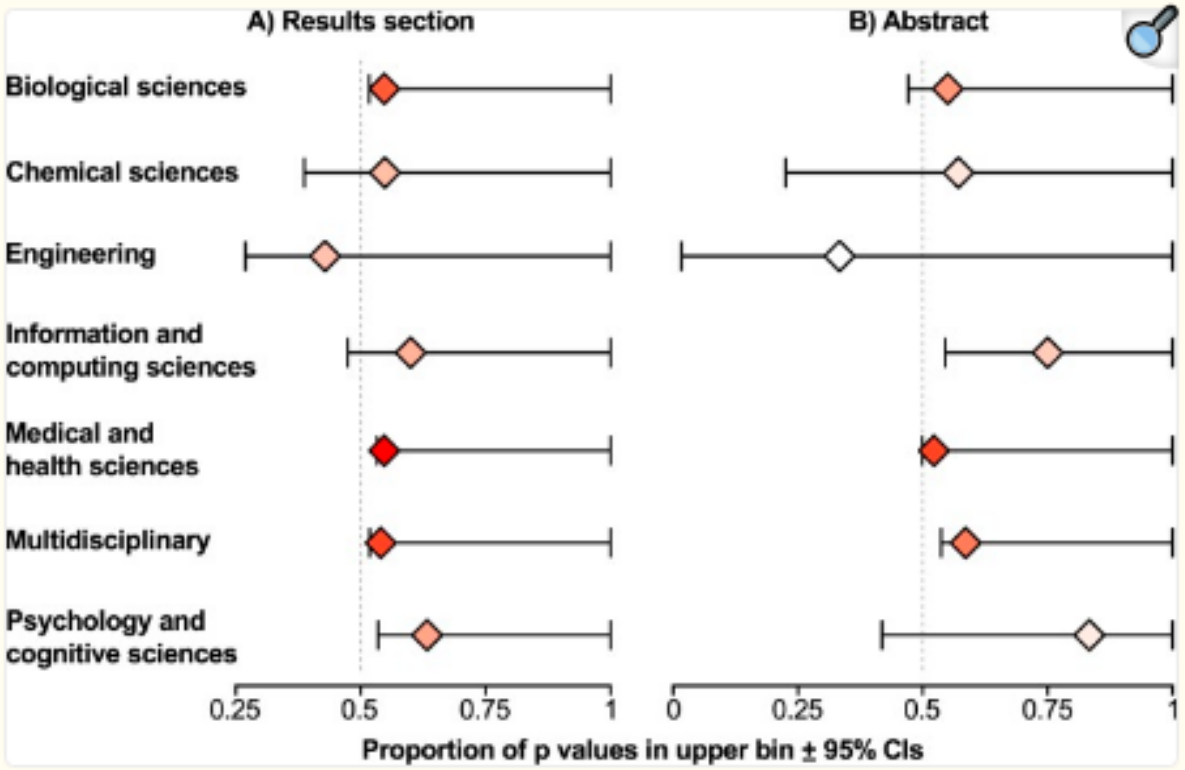
What is p-hacking?

Occurs when researchers collect or select data or statistical analyses until nonsignificant results become significant.

- **Optional Stopping:** Peeking at data and stopping when significant.
- **Covariate Fishing:** Trying different controls until *p* drops.
- **Outlier Manipulation:** Removing points selectively to help the Null fail.

> *Publication bias resulting from so-called "p - hacking" is pervasive throughout the life sciences; however, its effects on general conclusions made from the literature appear to be weak.*

Fig 3. Evidence for p-hacking across scientific disciplines.



Reference: Head et al. (2015) *The Extent and Consequences of P-Hacking in Science*.

Case Study: The Fisher-Mendel Controversy

Statistical Science
2010, Vol. 25, No. 4, 545–565
DOI: 10.1214/10-STS342
© Institute of Mathematical Statistics, 2010

A Statistical Model to Explain the Mendel–Fisher Controversy

Ana M. Pires and João A. Branco

1866

Mendel publishes "Experiments on Plant Hybridization".

1900

Rediscovery by de Vries, Correns, and von Tschermak.

1936

R.A. Fisher publishes his chi-square analysis.

The Claim

"Too good to be true."

3 citations in the next 35 years

Has Mendel's work been rediscovered? (Fisher, 1936)

Mendel's Experiment

The Setup

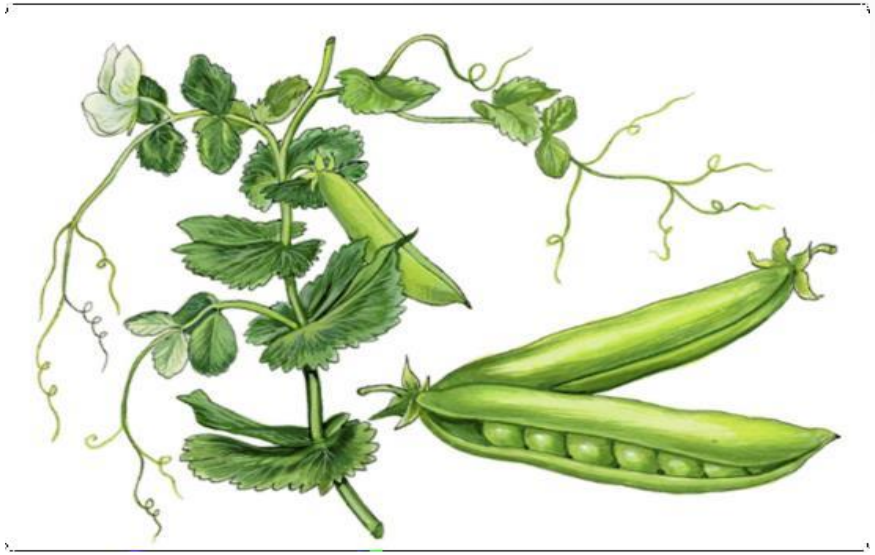
- Produced around 29,000 garden peas.
- **Single-trait:** 3:1 ratios (e.g., round vs. wrinkled).
- **Bifactorial:** e.g. Seed shape (A) and color (B).
- **Trifactorial:** Shape, color, and flower color.

He inferred egg and pollen composition based on gametic ratios.

TABLE 2

Data from the bifactorial experiment [as organized by Fisher (1936)]

	AA	*Aa*	*aa*	Total
BB	38	60	28	126
Bb	65	138	68	271
bb	35	67	30	132
Total	138	265	126	529



Source: www.thoughtco.com

TABLE 1

Data given in Mendel (1866) for the single trait experiments. "A" ("a") denotes the dominant (recessive) phenotype; A (a) denotes the dominant (recessive) allele; n is the total number of observations per experiment (that is, seeds for the seed trait experiments and plants otherwise); $n_{\text{"A"}}$, $n_{\text{"a"}}$, n_{Aa} and n_{AA} denote observed frequencies

	Trait	"A"	"a"	n	Obs. freq. $n_{\text{"A"}}$	Obs. freq. $n_{\text{"a"}}$	Theor. ratio "A" : "a"
	Seed shape	round	wrinkled	7324	5474	1850	3 : 1
	Seed color	yellow	green	8023	6022	2001	3 : 1
	Flower color	purple	white	929	705	224	3 : 1
F_2	Pod shape	inflated	constricted	1181	882	299	3 : 1
	Pod color	yellow	green	580	428	152	3 : 1
	Flower position	axial	terminal	858	651	207	3 : 1
	Stem length	long	short	1064	787	277	3 : 1
	Trait	***A***	***a***	***n***	n_{Aa}	n_{AA}	***Aa* : *AA***
	Seed shape	round	wrinkled	565	372	193	2 : 1
	Seed color	yellow	green	519	353	166	2 : 1
	Flower color	purple	white	100	64	36	2 : 1
(F_3)	Pod shape	inflated	constricted	100	71	29	2 : 1
	Pod color	yellow	green	100	60	40	2 : 1
	Flower position	axial	terminal	100	67	33	2 : 1
	Stem length	long	short	100	72	28	2 : 1
	Pod color (rep.)	yellow	green	100	65	35	2 : 1

Fisher's Chi-Square Analysis



Mendel's Data

Data adhered incredibly closely to the theoretical 3:1 ratio.

The Statistic

Aggregated P-value:

0.99993

Fisher suggested this implies unconscious or deliberate bias.



Fisher's View

Applied χ^2 goodness-of-fit. The probability of getting data *this* good is astronomically low.

Fisher's Chi-Square Analysis

- Aggregated p-value by summing the chi-square statistics & degrees of freedom
- Later simulations confirmed the anomaly

TABLE 5
Fisher's chi-square analysis ("Deviations expected and observed in all experiments")

Experiments	Expectation	χ^2	Probability of exceeding deviations observed
3 : 1 ratios	7	2.1389	0.95
2 : 1 ratios	8	5.1733	0.74
Bifactorial	8	2.8110	0.94
Gametic ratios	15	3.6730	0.9987
Trifactorial	26	15.3224	0.95
Total	64	29.1186	0.99987
Illustrations of plant variation	20	12.4870	0.90
Total	84	41.6056	0.99993

Analyses of p-values

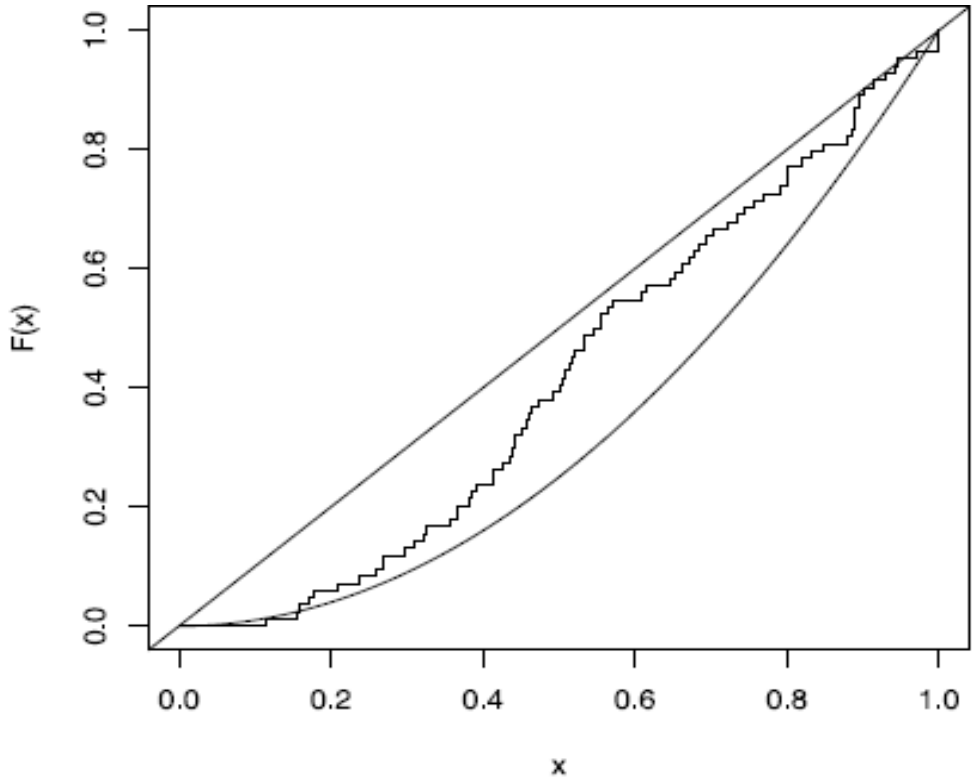


FIG. 6. *Empirical cumulative distribution function of the p-values (stair steps line); cumulative distribution function of the uniform (0, 1) random variable (straight line); cumulative distribution function of the maximum of two (0, 1) uniform random variables (curve).*

Under true nulls, p-values ~ Uniform(0,1)

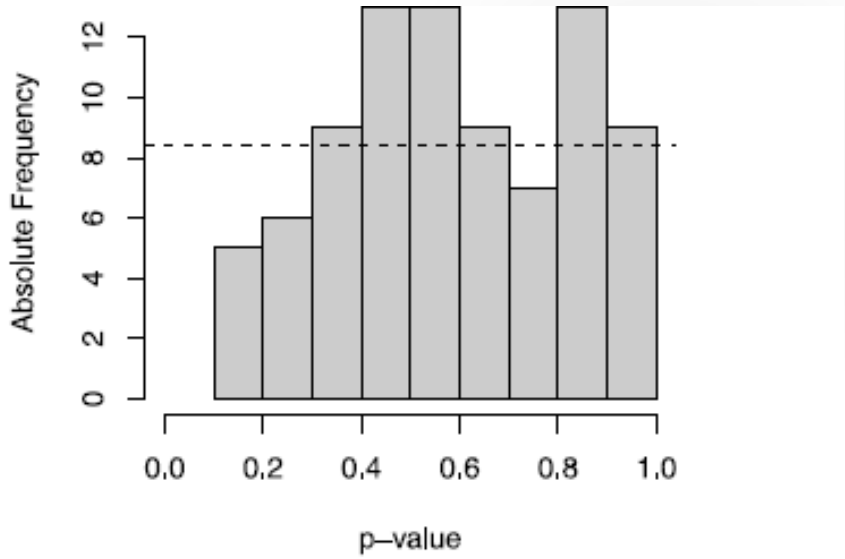


FIG. 7. *Histogram of the 84 p-values observed [the dashed line indicates the expected frequencies under the uniform (0, 1) distribution].*

Why? A Plausible Explanation: Censoring

The "Censoring" Hypothesis

Under true nulls, p-values must be Uniform(0,1).

However, if Mendel (or his assistants) repeated experiments when the results were "bad" $(p < \alpha)$and only reported the "good" ones, the distribution changes.

This creates a bias towards p-values close to 1.

```
If (p_value < 0.10) {
Repeat_Experiment();
} else {
Report_Data();
}
```

Result: Truncated Distribution

A plausible explanation

Assume Mendel repeated experiments when $p < \alpha$, reporting the higher max p

$$F_\alpha(x) = \begin{cases} x^2, & \text{if } 0 \le x \le \alpha, \\ (1+\alpha)x - \alpha, & \text{if } \alpha < x \le 1. \end{cases}$$

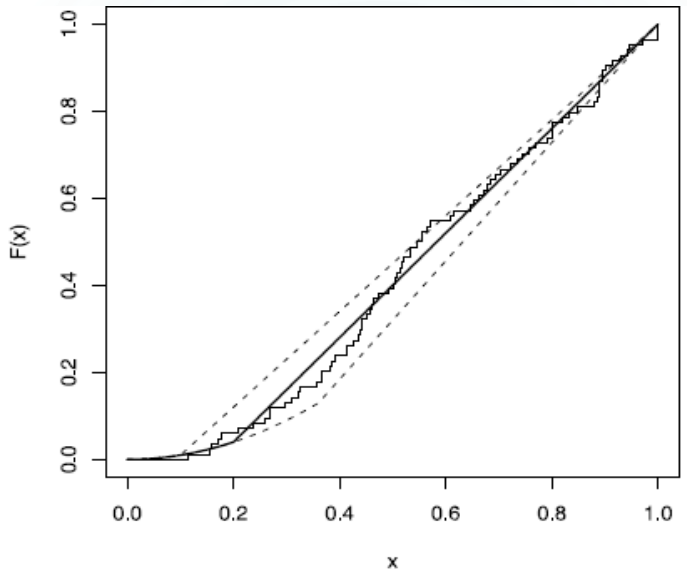


FIG. 9. *Empirical cumulative distribution function of the p-values and fitted model (solid line: $\hat{\alpha} = 0.201$; dashed lines: 90% confidence limits).*

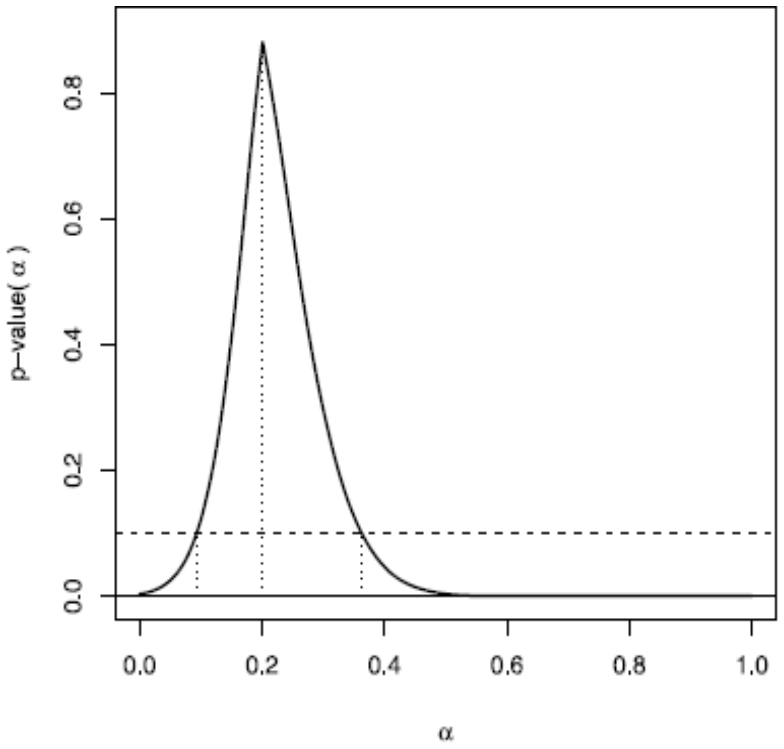


FIG. 8. *Plot of the p-value of the K–S test as a function of the parameter, showing the point estimate and the 90% confidence interval, for model A.*

Estimated $\alpha \approx 0.20$ (CI: 0.09 – 0.36)

Verdict & Further Support

Statistical Estimation

Modern re-analysis suggests an estimated censoring threshold of:

$$\alpha \approx 0.20$$

(CI: 0.09–0.36)

Mendel's Own Notes

Mendel was aware of potential bias. He wrote:

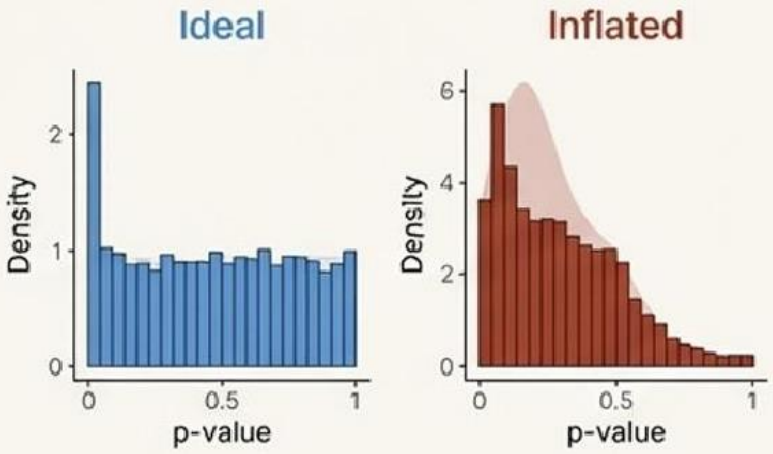
"...it appears to be necessary that all members of the series... should be, without exception, subjected to observation"

He mentions repeating an experiment where $p = 0.157$, hinting at this selective process.

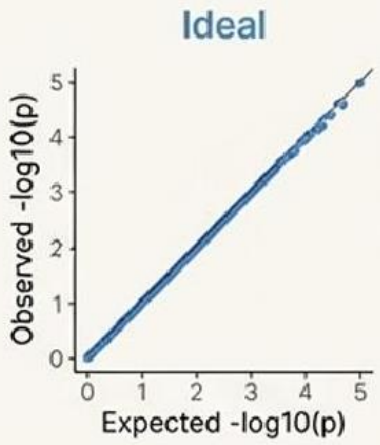
Diagnostic Your P-value Distribution

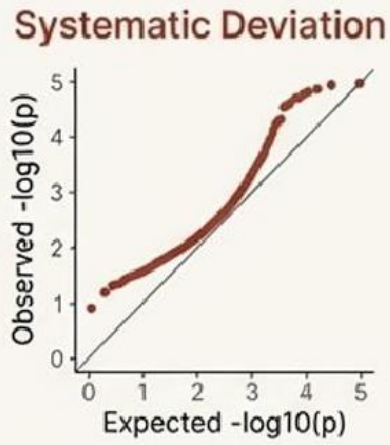
Understand your data: distribution of p-values

1. P-value Histogram: Is it uniform under the null?



2. Q-Q Plot: Do observed p-values follow the expected uniform distribution?





3. Genomic Inflation Factor (λ_{GC}): Is there systematic inflation?





Suspect model misspecification if λ_{GC} is high.

Credit:NotebookLm

Recommendations for Robust Analysis

I. **Make your test and Null hypothesis Specific**

II. **Choose the right criteria: FDR or FWER?**

III. **Careful Interpretation**

IV. **Always Visualize**

V. **Validate Independently**